BLOCK 8

PROBABILITY AND PROBABILITY DISTRIBUTIONS

UNIT 25 PROBABILITY THEORY

Structure

25.0 OBJECTIVES

After going through this unit, you should be able to

- identify the reasons for taking decisions under uncertain situations;
- define the concept of probability;
- differentiate between total probability and compound probability;
- describe the concept of independent events; and
- solve different probability problems in accordance with theoretical prescriptions.

25.1 INTRODUCTION

Probability theory is a branch of mathematics that is concerned with random (or chance) phenomenon. It originated in the games related to chance and an Italian mathematician Jerome Cardan was the first to write on the subject. However, the basic mathematical and formal foundation on the subject was provided by Pascal and Fermat. Contribution of Russian as well as European mathematicians helped the subject to grow.

In this unit, we will learn about some basic terminologies which are used in probability related matters, deterministic and non-deterministic experiments. Further we dwell into the theorems of probability (total and compound), and conditional probability. Before moving on to the Bayes' theorem, we will understand about the concept of independence and independent events.

25.2 DETERMINISTIC AND NON-DETERMINISTIC EXPERIMENTS

The agreement among scientists regarding the validity of most scientific theories rests, to a considerable extent, on the fact that the experiment on which the theory is based will yield the same result when they are repeated. If the same results are obtained when the experiment is repeated under the same conditions, we can conclude that the results are determined by the conditions, or the experiment is deterministic. For example, anywhere in the world if we throw a stone to the sky it will certainly come back to the earth after sometime.

However, there are experiments, which do not yield the same result even if the experimental conditions are kept constant. Examples are throwing a dice or picking a card from a well-shuffled pack of cards or tossing a coin. These experiments could be thought of as the ones with unpredictable results. If we are willing to stretch our idea of experiment, then there are many examples of this kind in our day-to-day life. For example, two people living in same conditions will die at different and unpredictable ages. In literature, these experiments or events are referred to as "random experiments" or "random events". Thus, we define "non-deterministic" or "random experiments" as those *experiments* (experiment is an act which can be repeated under some given conditions) whose outcomes are not predictable before-hand. Probability and branches of statistics are developed specially to deal with this kind of random events.

Each of the following may be called a random experiment:

1) Tossing a coin (or several coins)
2) Throwing a die (or several dice)
3) Drawing cards from a pack
4) Studying the distribution of boys and girls in families of India having two children

5) Drawing balls from an urn (or urns) having a given number of – different-kind of balls in each.

The result of a non-deterministic event is not predictable beforehand, for there may be a number of outcomes associated with it. The same outcome of a random experiment could be described in several ways. In our 2^{nd} example of random experiment, i.e., throwing a die, the possible outcomes are (1,2,3,4,5,6) These outcomes could have been described as “odd number of points” or “even number of points”.

The term event in probability theory is used to denote any phenomenon, which occurs as a result of a random experiment. Events can be ‘elementary’ as well as ‘composite’. An ‘elementary’ event cannot be decomposed into simpler events whereas a ‘composite’ event is an aggregate of several elementary events.

25.3 SOME IMPORTANT TERMINOLOGY

Sample Space: All possible outcomes of an on-deterministic experiment constitute the sample space of that experiment and are generally denoted by ‘S’.

For example, if one coin is tossed, either head(H)or tail (T)will appear. Therefore, H and T constitute the sample space and we can write S={H,T}.

Similarly if two coins are tossed, S={(HH),(HT),(TH),(TT)}

A subset of sample space, which might be favourable to the occurrence of a particular event, is called a sub-sample space. For example, in the previous example, S_1 = {(HH), (HT), (TH)} is a sub sample space and it is favourable to the event that atleast one head appears if two coins are tossed.

Example: The sample space of rolling two dice (x, y) of an experiment can be written as follows:

The number of events N(S) = 6 × 6 = 36:

$S = \{x, y\} : x = 1,2,3,\ldots,6; y = 1,2,3,\ldots,6$

Here x and y represent the number turned up in the top of the first and second dice respectively.

Events: When some of the elements of all the possible outcomes in the sample space of a random experiment satisfy a particular criterion, we call it an event. If three unbiased coins are tossed, the sample space is

S={(HHH),(HHT),(HTH),(HTT),(THH),(THT),(TTH),(TTT)}and

Following events can be picked up from it:

E_1= {(HHH)} = Event of getting all heads

E_2= {(HHH),(HHT),(HTH),(THH)}=Event of getting atleast 2 heads

E_3= {(HHT),(HTH),(THH)}=Event of getting exactly 2 heads…..……etc.

Example: Consider an event A that the total number of points rolled with a pair of dice is 7. So among the 36 possibilities only (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) yield a total of 7. So we write:

A = {(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)}

Thus an event (outcome or result) can be identified as a collection of points which constitute a subset of appropriate sample space for which the particular event occurs.

Mutually Exclusive Events: Events are said to be mutually exclusive if two or more of them cannot occur simultaneously. For example, while tossing a coin, the elementary events 'head' and 'tail' are mutually exclusive. Similarly, when we throw a die the appearance of the numbers 1, 2, 3, 4, 5, 6 are mutually exclusive. We define the events from these experiments as follows:

A: the event of odd number of points; and

B: the event of even number of points.

These are also mutually exclusive

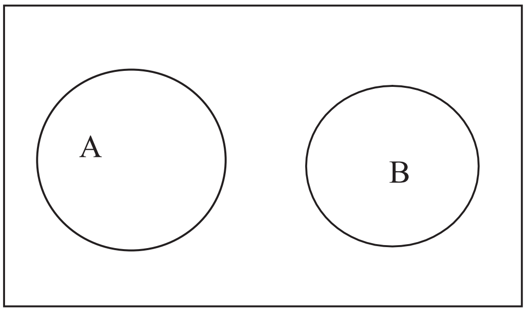

Fig. 25.1: Mutually Exclusive Events

In Fig. 25.1, the rectangular box represents the sample space and the circles represented by A and B represent subsample spaces and contain favourable elements to the events A and B. The complete separation of the circles indicates that there is no element, which is common to both the events. Thus, A and B events are mutually exclusive. The above way of representing events is called Venn diagram.

Mutually Exhaustive Events: Several events are said to be mutually exhaustive if and only if at least one of them necessarily occurs. For example, while tossing a coin, the events of head and tail are mutually exhaustive, as one of them must occur.

Equally Likely Events: The outcomes of a non-deterministic event are said to be equally Likely if occurrence of none of them can be expected in preference to another. For example, while tossing a coin, the occurrence of the event head or tail is equally likely if the coin is unbiased.

Exhaustive Events: Several events are said to form an exhaustive set, if atleast one of them must necessarily occur. The complete group of all elementary events of a random experiment given is an exhaustive set of events. For example, in tossing a coin the two events head or tail form an exhaustive set because one of these two must necessarily occur.

Independent Events: Events are said to be independent of each other if occurrence of one event is not affected by the occurrence of the others. For example, while throwing a die repeatedly, the event of getting a '3' in the first throw is independent of getting a '6' in the second throw.

Conditional Events: When events are neither independent nor mutually exclusive, it is possible to think that one of them is dependent on the other. For example, it may or may not rain if the day is cloudy but if there is rain, there must be clouds in the sky. Thus, the event of rain is conditioned upon the event of clouds in the sky.

25.4 DEFINITIONS OF PROBABILITY

The term probability has been interpreted in terms of four definitions viz.,

1) Classical definition
2) Axiomatic definition
3) Empirical definition
4) Subjective definition

25.4.1 Classical Definition

The classical definition states that if an experiment consists of N outcomes which are mutually exclusive, exhaustive and equally likely and N_A are the number of outcomes that are favourable to an event A, then the probability of the event A (P(A))is defined as

$P(A)=N_A/N$

In other words, the probability of an event A equals the ratio of the number of outcomes N_A favourable to A to the total number of outcomes. See the following example for a better understanding of the concept.

Example: Two unbiased dice are thrown simultaneously. Find the probability that the product of the points appearing on the dice is18.

There are 36 (N) possible outcomes if two dice are thrown simultaneously. These outcomes are mutually exclusive, exhaustive and equally likely based on the assumption that the dice are unbiased. Now we denote A: the product of the points appearing on the dice is 18.

The events favourable to 'A' are [(3, 6), (6, 3)] only, therefore, $N_{A=}$ 2. According to classical definition of probability

$P(A)= N_A/N=1/18$

When none of the outcome is favourable to the event A, N_A= 0, P (A) also takes the value 0. In that case we say that event A is impossible. If $N_A = N, P(A)$ takes the value 1. In that case we say that event A is a sure event.

There are some limitations of the classical definition of probability. Unless the outcomes of an event are mutually exclusive, exhaustive and equally likely, classical definition cannot be applied. Again, if the number of outcomes of an event is infinitely large, the definition fails. The phrase 'equally likely' appearing in the classical definition of probability means equally probable, thus the definition is circular in nature.

Techniques of Counting

1) Fundamental Principle of Counting if several processes can be performed in the following manner: the first process in p ways, the second in q ways, the third in r ways, and so on, then the total number of ways in which the whole process can be performed in the order indicated is given by their product: p.q.r….

2) Permutation: The total number of ways of arranging n distinct objects taken r at a time is given by: ${}^nP_r$

   $${}^nP_r = n(n-1)(n-2)......(n-r+1)$$

3) Arrangement in a line or circle: The total number of ways in which n distinct objects can be arranged among themselves is:

   i) in a line: $n! = 1.2.3.4....n$

   ii) in a circle: $(n - 1)!$

4) Permutation with repetition: The number of ways of arranging n objects among which p are alike, q are alike, r are alike, etc. is: $\dfrac{n!}{p!.q!.r!.....}$

5) Combination: The total number of possible groups that can be formed by taking r objects out of n distinct objects is given by: ${}^nC_r =$

   $${}^nC_r = \frac{n!}{r!(n-r)!} = \frac{n(n-1)(n-2)...(n-r+1)}{r!}$$

6) Choosing balls from an urn: The total number of ways of choosing a number of white balls and b number of black balls from an urn containing A number of white balls and B number of black balls is: ${}^AC_a . {}^BC_b$

7) Ordered Partition (Distinct Objects): The total number of ways of distributing n distinct objects into r components marked as 1, 2, 3, 4, …, r is r^n

   The number of ways in which n objects can be distributed so that the r compartments contain respectively $n_1, n_2, n_3, ..., n_r$ objects is $\dfrac{n!}{n_1!n_2!n_3!..., n_r!}$

8) Ordered Partition (identical objects): The total number of ways of distributing n identical objects into r compartments marked as 1, 2, 3, …r is ${}^{n+r-1}C_{r-1}$

If none of the compartments should remain empty, the total number of ways of distributing the objects is ${}^{n-1}C_{r-1}$

25.4.2 Axiomatic Definition

In the axiomatic definition of probability, we start with a probability space 'S' where set 'S' of abstract objects is called outcomes. The set S and its subsets are called events. The probability of an outcome A is by definition a number P (A) assigned to A. Such a number satisfies the following axioms:

a) $P(A) \geq 0$ i.e., P(A) is a non-negative number.

b) The probability of the certain event S is 1, i.e., P(S)=1.

c) If two events A and B have no common elements, or, A and B are mutually exclusive, the probability of the event consisting of the outcomes that are in A or in B equals to sum of their probabilities, which can be written as A union B (AUB):

   P(AUB)=P(A)+P(B)

We comment next on the connection between an abstract sample space and the underlying real experiment. The first step in model formation is between elements of S and its experimental outcomes. The actual outcomes of a real experiment can involve a large number of observable characteristics. In the formation of the model, we select from these characteristics the one that is of interest in our investigation.

For example, consider the possible models of the throwing of an unbiased die by the 3 players X, Y and Z.

X says that the outcomes of this consist of six faces of the die, forming the sample space{1,2,3,4,5,6}.

Y argues that the experiment has only 2 outcomes, even or odd, forming the sample space {even, odd}

Z bets that the die will rest on the left side of the table and the face with one point will show. Her experiment consists of infinitely many points consisting of the six faces of the die and the coordinate of the table where the die rests finally.

25.4.3 Empirical Definition

In N trials of a random experiment, if an event is found to occur m times, the relative frequency of the occurrence of the event is m/N. If this relative frequency approaches a limiting value p, as N increases indefinitely, then 'p' is called the probability of the event A.

$$P(A) = \lim_{N \to \infty} \left(\frac{m}{N} \right)$$

To give a meaning to the limit we must interpret the above formula as an assumption used to define P(A). This concept was introduced by Von Mises. However, the use of such a definition as a basis of deductive theory has not enjoyed wide acceptance.

25.4.4 Subjective Definition

In subjective interpretation of probability, the number P (A) is assigned to a statement. A, is a measure of our state of knowledge or belief concerning the truth of A. These kinds of probabilities are most often used in our daily life and conversations. We often make statements like "I am 100% sure that I will pass the examination" i.e., P(of passing the examinations) = 1, or "there is 50% chance that India will win the match against Pakistan" i.e., P(India will win the match against Pakistan) = ½

Check Your Progress 1

1) What is the probability that all three children born in a family will have different birthdays? Hint: Use rule number 1 (Techniques of Counting section)

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2) Five persons a, b, c, d, e occupy seats in a row at random. What is the probability that a and b will sit next to each other? Hint: Use rule number 3 (Techniques of Counting section)

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3) Two cards are drawn at random from a pack of well-shuffled cards. Find the probability that

   a) Both cards are red

   b) One is a heart and another is a diamond

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4) A bag contains 6 white and 4 red balls. One ball is drawn at random. What is the probability that it will be white?

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5) 15 identical balls are distributed at random into 4 boxes numbered 1, 2, 3, 4. Find the probability that, (a) each box contains at least 2 objects and (b) no box is empty.

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6) A box contains 20 identical tickets, the tickets being numbered as 1,2,3,…., 20. If 3 tickets are chosen at random, what is the probability that the numbers on the drawn tickets will be in arithmetic progression?

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25.5 THEOREMS OF PROBABILITY

In this section, we will consider the basic theorems of probability. We will denote the sub-sample spaces by A,B,C … which represent the elements of the sample space S, which are favourable to the events A, B and C.

Note that P(S)=1 and P(A),P(B),P(C)….liebetween0 and 1.

25.5.1 Theorem of Total Probability

If two events A and B are mutually exclusive, exhaustive and equally likely, then the probability of occurrence of either A or B, is given by P(A + B) or P(A U B) is given by the sum of their probability. Thus,

P(AU B) = P(A)+ P(B)

This is also known as the Addition Theorem.

Let us assume that a random experiment has n possible outcomes which are mutually exclusive, exhaustive and equally likely. While m_1 of them are favourable to A, m_2 are favourable to B. By the classical definition of probability

P(A) = m_1/nandP(B) = m_2/n

Since A and B are mutually exclusive and exhaustive, the number of m_1 cases favourable to the event A is completely distinct from m_2 cases favourable to the event B. Thus, the number of cases favourable to A and B are (AUB), is therefore given by $m_{1+}m_2$,

$P(A+B) = P(AUB)=(m_1+m_2)/n=(m_1/n)+(m_2/n)=P(A)+P(B)$

Deductions from Theorem of Total Probability

1) **Theorem of Complementary Event**

If A denotes the occurrence of the event A, then A^c (read as 'compliment of A') denotes non-occurrence of the event A and

$P(A) = 1 - P(A^c)$.

Since A and A^c are mutually exclusive and exhaustive events, $S = \{A, A^c\}$. Applying the theorem of total probability we get,

$P(S) = P(A) + P(A^c) = 1$

or, $P(A^c) = 1 - P(A)$

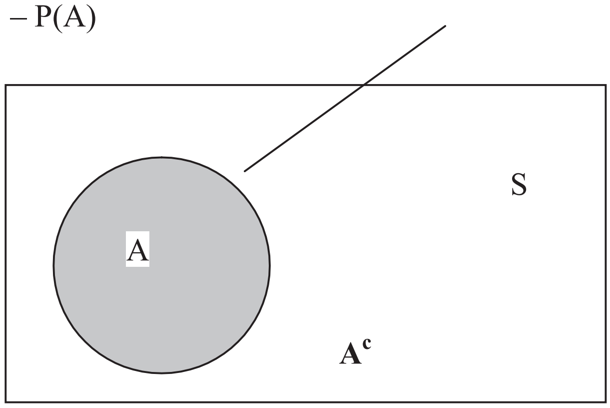

Fig. 25.2: Complementary Event

See that this theorem is very intuitive. If the probability of getting head while tossing an unbiased coin is 5, then the probability of getting a tail is obviously 0.5(1–.5).

2) **Extension of Total Probability Theorem**

The theorem of total probability can be extended to any number of mutually exclusive events. If the events $A_1, A_2, A_3, \ldots\ldots, A_k$ are mutually exclusive, then the probability of occurrence of any one of them ($U^k{}_{i=1}\ A_i$) is given by the sum of their probabilities.

$P(U^k{}_{i=1}A_i) = P(A_1 + A_2 + A_3 + \ldots,+A_k) = P(A_1)+P(A_2)+P(A_3)+\ldots\ldots\ldots.+P(A_k)$

3) **Theorem of Total Probability with Mutually Non-exclusive Events**

The probability of occurrence of at least one of the events A and B (which are not necessarily mutually exclusive) is given by

$P(AUB) = P(A) + P(B) - P(A\cap B)$

The symbol '∩' means joint occurrence or 'and'. Hence, (A ∩ B) means the occurrence of the event A and B, whereas 'U' means 'or' i.e., (A U B)$\Rightarrow$ the occurrence of either the event A or the event B.

Proof: The occurrence of the event (A U B) is analogous to the occurrence of any one of the following three mutually exclusive events:

$(A\cap B^c)$, $(A^c\cap B)$ and $(A\cap B)$. i.e.,

$$P(A+B)=P(A\cup B)=P(A\cap B^c)+(A^c+\cap B)+(A\cap B)=P(A\bar{B})+P(\bar{A}B)+P(AB)$$

In terms of Venn diagram,

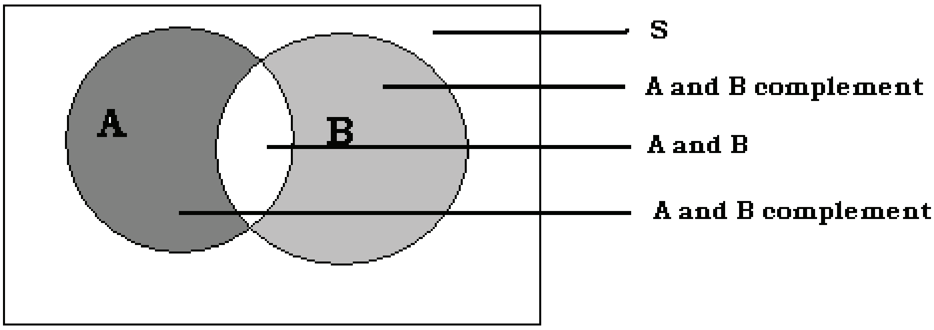

Fig. 25.3: Union and intersection of A & B

Therefore, using the definition of total probability, we get

$P(AUB) = P(A + B) = P(A\cap B^c) + P(A^c\cap B) + P(A\cap B) = P(A\bar{B})+P(\bar{A}B)+P(AB)$ ……….[1]

Again, the occurrence of A is analogous to the occurrence of any one of the following two mutually exclusive events P (A ∩ B) i.e., P(AB) and $P(A \cap B^c)$ i.e., $P(A\bar{B})$, Thus, we get

$P(A) = P(A\cap B) + P(A\cap B^c)$ or $P(A)=P(A\bar{B})+P(AB)$ ……[2]

Similarly, for B

$P(B) = P(B\cap A) + P(B\cap A^c)$ or $P(B)=P(AB)+(\bar{A}B)$ ……[3]

Using [1], [2], [3] we can derive that

P(AUB) = P(A + B) = P(A) – P(AB) + P(B) – P(AB) = P(A) +P(B) – P(AB) = P(A) + P(B) – P $(A\cap B)$.

The above result could be extended to three events A, B, C, which are not mutually exclusive

$P(AUBUC) = P(A) + P(B) + P(C) – P(A\cap B) – P(A\cap C) – P(C\cap B) + P(A\cap B\cap C)$

We illustrate the situation in Fig. 25.4.

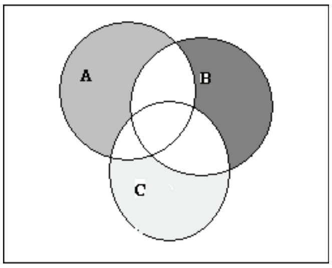

Fig. 25.4: Union and intersection of A, B and C

In this context, we mention that there are two standard results in the theory of probability

1) $P(AUB) \leq P(A) + P(B)$ .... Boole's inequality
2) $P(A \cap B) \geq P(A) + P(B) - 1$ .... Bonferroni's inequality

25.5.2 Theorem of Compound Probability

The probability of occurrence of the event A and B is given by the product of the probability of the event A and *conditional probability* of the event B given that A has actually occurred, which is denoted by P(A|B). P(A|B) is given by the ratio of the number of events favourable to the event A and B to the number of events favourable to the event A. Symbolically,

$P(A \cap B) = P(A) \times P(B|A)$.

Suppose a random experiment has n mutually exclusive, exhaustive and equally likely outcomes among which $m_{1,}$ m_2 and m_{12} are favourable to the events A, B and $(A \cap B)$ respectively.

So, $P(A) = m_1/n$;

The number of cases favourable to A as well as B: $P(A \cap B)=m_{12}/n$

Note that event A has actually occurred, and so the occurrence of the event B is limited to only m_1 cases. Thus $P(B/A) = m_{12}/m_1$. Therefore,

$P(AB)\ P(BA) = m_1/n \times m_{12}/m_1 = m_{12}/n = P(AB)$

This theorem is also known as the multiplication theorem.

25.5.2.1 Deductions from Theorem of Compound Probability

The occurrence of one event, say, B may be associated with the occurrence or non-occurrence of another events say, A. This in turn implies that we can think of B to be composed of two mutually exclusive events $(A \cap B)$ and $(A^c \cap B)$. Applying the theorem of total probability

$P(B)=P(A \cap B)+P(A^c \cap B)$

$=P(A) \times P(B|A)+P(A^c) \times P(B|A^c)$.........[using theorem of compound probability]

1) **Extension of Compound Probability Theorem**

The above theorem can be extended to include the cases when there are three or more events. Suppose there are three events A,B and C. Then

$P(A\cap B\cap C) = P(A) \times P(B|A) \times P(C/(A\cap B)$

And soon for more than three events.

Example: Given $P(A) = 1/4$, $P(B) = 2/5$ and $P(A \cup B) = ½$ find $P(A|B)$ and $P(B|A)$.

$P(A\cap B) = P(A) + P(B) – P(A\cup B) = ¼ + 2/5 – ½ = 3/20$

Therefore, $P(A/B) = P(A\cap B) / P(B) = 3/20 \times 5/2 = 3/8$ and $P(B/A) = P(A\cap B) / P(A) = 3/20 \times 4 = 3/5$

Example: At an examination in three courses A, B and C the following results were obtained

25% of the candidates passed in course A

20% of the candidates passed in course B

35% of the candidates passed in course C

7% of the candidates passed in course A and B

5% of the candidates passed in course A and C

2% of the candidates passed in course B and C

1% of the candidates passed in all the subjects.

Find the probability that a candidate got pass marks in atleast one course.

$P(A) = 0.25$, $P(B) = 0.2$, $P(C) = 0.35$, $P(A\cap B) = 0.07$, $P(C\cap B) = 0.02$, $P(A\cap C) = 0.05$ and $P(A\cap B\cap C) = 0.01$.

Therefore, $P(A\cup B \cup C) = .25 + .2 + .35 - .07 - .05 - .02 + .01 = .67$

Check Your Progress 2

1) If $P(A) = ½$, $P(B)=1/3$, $P(A\cap B)=¼$ find $P(A^c)$, $P(A\cup B)$, $P(A|B)$, $P(A^c\cap B)$, $P(A^c\cap B^c)$, $P(A^c\cup B)$. State whether the events A and B are mutually exclusive, equally likely, exhaustive.

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2) For three events A, B, C which are not mutually exclusive, prove $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A\cap B) - P(A\cap C) - P(C\cap B) + P(A\cap B\cap C)$ usingVenn diagram.

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25.6 CONDITIONAL PROBABILITY AND CONCEPT OF INDEPENDENCE

25.6.1 Conditional Probability

From the theorem of compound probability we can get the probability of one event, say, event B conditioned on some other event, say A. As we have discussed earlier, this is symbolically written as $P(B|A)$. From the theory of compound probability, we know that

$P(A\cap B) = P(A)\times P(B|A)$

or, $P(B|A) = P(A\cap B)/P(A)$ provided that $P(A)\neq 0$.

Example: Find out the probability of getting the Ace of hearts when one card is drawn from a well-shuffled pack of cards given the fact that the card is red.

Let A denote the event that the cardis red and B denote the event that the card is the Ace of hearts. Then we are interested in finding P (B|A). From the theorem of conditional probability

$P(B|A) = P(A\cap B)/P(A) = (13/52)\,(1/13) / (26/52) = (1/52)\,(52/26) = 1/26$

25.6.2 Concept of Independent Events

Two events A and B are said to be statistically independent if the occurrence of one event is not affected by the occurrence of another event. Similarly, several events are said to be independent, if the occurrence of one event is not affected by the supplementary knowledge of the occurrence of other events. This implies

$P(B|A) = P(B|A^c) = P(B)$

Therefore, from the theorem of compound probability, we get

$P(A\cap B) = P(A) \times P(B|A)$

$= P(A) \times P(B)$

Similarly, for three events that are mutually or statistically independent, we have the following results

$P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$ along with $P(A\cap B)=P(A)\times P(B)$

$P(C \cap B) = P(C) \times P(B)$

$P(C \cap A) = P(C) \times P(A)$

For more events A, B, C, D to be mutually independent following should hold:
$P(A \cap B \cap C \cap D) = P(A) \times P(B) \times P(C) \times P(D)$ along with

$P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$

$P(A \cap B \cap D) = P(A) \times P(B) \times P(D)$

$P(D \cap B \cap C) = P(D) \times P(B) \times P(C)$

$P(A \cap D \cap C) = P(A) \times P(D) \times P(C)$

$P(A \cap B) = P(A) \times P(B)$

$P(C \cap B) = P(C) \times P(B)$

$P(C \cap A) = P(C) \times P(A)$

And soon……..

Thus, two events are said to be independent if the probability of occurrence of both or all equals the product of their probabilities.

For pairwise independence of events the above should hold for any two of the events.

Deductions from the Concept of Independence

If the events A and B are independent then A^c and B^c are also independent.

Since A and B are independent

$P(A \cap B) = P(A) \times P(B)$

$P(A^c \cap B^c) = P(AUB)^c$……[De Morgan's theorem]

$= 1 - P(AUB)$

$= 1 - P(A) - P(B) + P(A \cap B)$

$= 1 - P(A) - P(B) + P(A) \times P(B)$ $\qquad [\because P(AUB) = P(A) + P(B) - P(A \cap B)]$

$= \{1 - P(A)\}\{1 - P(B)\}$

$= P(A^C) \times P(B^C)$

Example: Given that $P(A)=3/8$ and $P(B)=5/8$ and $P(AUB)=3/4$, find $P(B|A)$ and $P(A|B)$. Are A and B independent?

Using the relationship

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$, we get $P(A \cap B) = 1/4$

Thus, the given information does not satisfy the equation $P(A \cap B) = P(A) \times P(B)$

Therefore, A and B are not independent.

Example: One urn contains 2 white and 2 red balls and a second urn contains 2 white and 4 red balls

a) If one ball is selected from each urn, what is the probability that they will be of the same colour?

b) If an urn is selected at random and then a ball is selected at random, what is the probability that it will be a white ball?

Ans.

a) Let A denote the event that both the balls drawn from each urn are of the same color. A_1 denotes that they are white and A_2 denotes that they are red. Clearly, A_1 and A_2 are two mutually exclusive events. Applying the theorem of total probability,

$P(A)=P(A_1)+P(A_2)$

Here A_1 is a compound event formed by two independent events of drawing a white ball from each urn, Therefore, $P(A_1) = ½ \times 1/3 = 1/6$. Similarly, $P(A_2) = \frac{1}{2} \times \frac{2}{3} \times = \frac{1}{3}$

Hence, $P(A) = 1/6 + 2/6 = ½$.

b) A white ball can be selected in two mutually exclusive ways, when urn 1 is selected and a white ball is drawn from it (denoted by the event A) and when urn 2 is selected and a white ball is drawn from it (denoted by the event B).

P(A)=P(urn1is selected)×P(a white ball is drawn),because selection of urn 1 and selection of a white ball are mutually independent.

$P(A) = ½ \times ½ = ¼$.

Similarly,

$P(B) = ½ \times 1/3 = 1/6$

Using the theorem of total probability,

P(drawing a white ball from an urn)=P(A)+ P(B)=5/12.

Check Your Progress 3

1) A salesman has 50% chance of making a sale. If two customers enter the shop, what is the probability that the salesman will make a sale?

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2) If the events A and B are independent, then show that Ac,Bc,A and B are pair wise independent.

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3) If A and B are mutually exclusive events, then show that P(A|AUB) =P(A)/P(A)+P(B).

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4) A bag contains 8 red and 3 white balls. Two successive draws of 3 balls are made without replacement. Find the probability that the first drawing will give 3 white balls and the second 3 red balls.

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25.7 BAYES' THEOREM AND ITS APPLICATION

Suppose an event A can occur if and only if one of the mutually exclusive events B_1, B_2, B_3,…………….., B_n occurs. If the unconditional probabilities $P(B_1)$,$P(B_2)$,$P(B_3)$,……………,$P(B_n)$ are known and the conditional probabilities are $P(A|B_1)$, $P(A|B_2)$, $P(A|B_3)$,………., $P(A|B_n)$ and known, then the conditional probability $P(B_i|A)$ could be calculated when A has actually occurred.

$$P(A)=\sum_{i=1}^{n}P(A\cap B_i)=\sum_{i=1}^{n}P(B_i)P(A\,|\,B_i)$$

$$P(B_i\,|\,A)=\frac{P(B_i\cap A)}{P(A)}=\frac{P(A\,|\,B_i)\times P(A)}{\sum_{i=1}^{n}P(B_i)P(A\,|\,B_i)}\text{, therefore}$$

$$P(B_i\,|\,A)=\frac{P(A\,|\,B_i)\times P(A)}{\sum_{i=1}^{n}P(B_i)P(A\,|\,B_i)}$$

This is known as Bayes' theorem. This is a very strong result in the theory of probability. An example will illustrate the theorem more vividly.

Example: Two boxes contain respectively 2 red and 2 black balls and 2 red and 4 black balls. One ball is transferred from one box to another and then one ball is selected from the second box. If it turns out to be black, what is the probability that the transferred ball was red?

B_1:the transferred ball was red.

B_2:the transferred ball was black.

A: the ball selected from the second box is black. $P(B_1)=½$

$P(B_2) = ½$

$P(A| B_1) = 3/7$

$P(A|B_2)=5/7$

$P(B_1|A)=P(B_1)\times P(A|B_1)/P(B_1)\times P(A|B_1)+P(B_2)\times P(A|B_2)$

$=½ \times 3/7/(½ \times 3/7+½\times 5/7)=3/8$

Check Your Progress 4

1) In a bulb factory there are three machines a, b and c. They produce 25%, 35% and 40% of total product. Of their output 5, 4 and 2 percent of bulbs respectively are defective. If one bulb is selected at random, what is the probability that it was produced by the machine?

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2) We have two coins. The first coin is fair with probability of head= ½, but the second coin is loaded with head. Therefore, probability of getting head in the second coin = 2/3. Suppose we pick one coin at random, we toss it, and the head shows. Find the probability that we picked the fair coin.

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25.8 MATHEMATICAL EXPECTATIONS

Suppose a random experiment has n mutually exclusive and exhaustive outcomes. Let the variable x take values $x_1, x_2, x_3, \ldots, x_n$ with probabilities $p_1, p_2, p_3, \ldots, p_n$. Then the mathematical expectation or the expected value of the variable is defined as the weighted sum of the values of the variable, the weights being the probabilities of the values of the variable. Expected value of a variable 'x' is denoted by E(x). Therefore,

$$E(X) = p_1x_1 + p_2x_2 + \ldots + p_nx_n$$

Provided that $\sum_{i=1}^{n} p_i = 1; p_i \geq 0$

If E(x)=m, the mathematical expectation of the variable$(x - m)^2$isknownas the variance of the variable x, which is denoted by Var(x).

$$Var(x) = E(x - m)^2 = (x_1 - m)^2 p_1 + (x_2 - m)^2 p_2 + (x_3 - m)^2 p_3 + \ldots +$$

$$(x_n - m)^2 p_n = \sum_{i=1}^{n} (x_i - m)^2 p_i.$$

We can show that $Var(x) = E(x - m)^2 = E(x^2) - [E(x)]^2$. Remember, in our notation, E(x) = m. The square root of Var(x) is called the standard deviation of the variable x.

If g(x) is any function of the variable x, defined for every value of the variable x, then the expected value of the function (x), denoted by E[g(x)] is given by

$$\Sigma g(x_i) p_i = g(x_1) p_1 + g(x_2) p_2 + g(x_3) p_3 + \ldots + g(x_n) p_n = \sum_{i=1}^{n} (X_i - m)^2 p_i$$

Mathematical expectation of a variable is analogous to weighted Arithmetic Mean of the variable, say, x. In a frequency distribution, the relative frequencies (Class frequency / Total frequency) of a variable could be thought of as the probability that the variable will take that value, i.e. $p_i=f_i/N$,

where p_i : Probability that the variable x will take the value x_i

f_i : frequency of occurrence of x_i

N: Total frequency $\sum_{i=1}^{n} f_i$

As we know weighted arithmetic mean $= \frac{\sum_{i=1}^{n} X_i f_i}{\sum_{i=1}^{n} f_i} = \frac{\sum_{i=1}^{n} X_i f_i}{N} = \sum_{i=1}^{n} X_i p_i = E(x)$

= Expected value of the variable x.

Example: What is the mathematical expectation of the number of points when a nun biased die is thrown?

Let the variable denote the number of points when a die is thrown. Therefore, *x*can take values 1, 2, 3, 4, 5 and 6. The probability of realization of all these values is same, viz.,1/6,therefore,

E(*x*)=1/6(1+2+3+4+5+6)=3.5

Theorem 1

The mathematical expectation of sum of several random variables is equal to the sum of the mathematical expectation of the random variables.

E(a+b+c+d+……….) = E(a)+E(b)+E(c)+E(d)+……., where a, b, c, d represent random variables.

We will consider the theorem for two random variables and the result could be extended to any number of random variables.

Let, x and y are two random variables; x can take the values $x_1, x_2, x_3,\ldots\ldots\ldots,x_n$ and y can take the values $y_1,y_2,y_3,\ldots\ldots\ldots,y_m,$ and p_{ij} is the probability of the event that $x = x_i$ and $y = y_j$. Now $(x + y)$ is a new random variable and it takes the value $(x_i + y_j)$ with probability p_{ij} or, $P(x = x_i \text{ and } y = y_j) = p_{ij}$. Using the definition of mathematical expectation

$E(x+y)=\sum_{i=1}^{n}\sum_{j=1}^{m}(x_i+y_i)p_{ij}$ [we take double summation as i can take n values and j can take m values]

$$E(x+y)=\sum_{i=1}^{n}\sum_{j=1}^{m}(x_ip_{ij}+y_jp_{ij})=\sum_{i=1}^{n}\sum_{j=1}^{m}x_ip_{ij}+\sum_{i=1}^{n}\sum_{i=1}^{m}y_jp_{ij}$$

$=\sum_{i=1}^{n}x_i\sum_{j=1}^{m}p_{ij}+\sum_{j=1}^{m}y_j\sum_{i=1}^{n}p_{ij}$ [we could write this as x_i is constant with respect to variations in j]

The following table will explain how we could write the above. In this context, we define marginal probability of a variable given the other variable takes some specific value. Suppose n= 9 and m= 8, i.e., the variables x and y take 9 and 8 values respectively. From the table, it is easy to see the distribution of the variables and their probabilities, where the symbols have their usual meanings.

Marginal probability of y takes the value $yj=\sum_{i=1}^{9}p_{i0}=\sum_{j=1}^{8}p_{oj}$

X ↓ \ y →	y1	y2	y3	y4	y5	y6	y7	y8	Marginal Probability of x ↓
x1	p11	p12	p13	p14	p15	p16	p17	p18	p10
x2	p21	p22	p23	p24	p25	p26	p27	p28	p20
x3	p31	p32	p33	p34	p35	p36	p37	p38	p30
x4	p41	p42	p43	p44	p45	p46	p47	p48	p40
x5	p51	p52	p53	p54	p55	p56	p57	p58	p50
x6	p61	p62	p63	p64	p65	p66	p67	p68	p60
x7	p71	p72	p73	p74	p75	p76	p77	p78	p70
x8	p81	p82	p83	p84	p85	p86	p87	p88	p80
x9	p91	p92	p93	p94	p95	p96	p97	p98	p90
Marginal Probability of y →	p01	p02	p03	p04	p05	p06	p07	p08	1

Similarly, marginal probability of x takes the value $x_i = \sum_{j=1}^{8} p_{ij} = p_{i0}$

In the above situation, $E(x) = \sum_{i=1}^{9} x_i p_{0i}$ and $E(y) = \sum_{i=1}^{n} y_i p_{j0}$

Additionally, we can also derive the conditional distribution of the variables from the above table. One example will elucidate it. The conditional distribution of x given $y = y_1$is as follows:

	$y = y_1$	Conditional Probability distribution of x given $y = y_1$
x_1	p_{11}	p11 / p01
x_2	p_{21}	p21 / p01
x_3	p_{31}	p31 / p01
x_4	p_{41}	p41 / p01
x5	p_{51}	p51 / p01
x6	p61	p61 / p01
x7	p71	p71 / p01
x8	p81	p81 / p01
x9	p91	p91 / p01
Total	p01	1

Therefore, $E(x+y) = \sum_{i=1}^{n} x_i \sum_{j=1}^{m} p_{ij} + \sum_{j=1}^{m} y_j \sum_{i=1}^{n} p_{ij}$

$$= \sum_{i=1}^{n} x_i p_{i0} + \sum_{j=1}^{m} y_j p_{0j} = E(x) + E(y)$$

Theorem2

The mathematical expectation of product of several independent random variables is equal to the product of the mathematical expectation of the random variables.

We retain the symbols of the previous theorem and additionally, we assume that the variables x and y are independent, i.e., occurrence of any one of the event has no impact on the occurrence of the other. Let the variable x take the value x_i with probability p_i and the variable y takes the value y_j with probability q_j .Since x and y are independent

$P(x = x_i \text{ and } y = y_j) = p_i \times q_j$.

The theorem states that

$E(x.y)= E(x)\times E(y)$.

Using the definition of mathematical expectation

$$E(x,y)=\sum_{i=1}^{n}\sum_{j=1}^{m}x_iy_jp(x=x_i \text{ and } y=y_j)=\sum_{i=1}^{n}\sum_{j=1}^{m}x_iy_j(p\times q_j)$$

Summing over j and keeping i constant

$$=\sum_{i=1}^{n}x_ip_i\times\sum_{j=1}^{m}y_jq_j)=E(x)\times E(y).$$

This theorem could be extended to any number of variables.

Corollary 1: If m is the expected value of the variable x, then

$$E(x-m)^2=E(x^2)-[E(x)]^2.$$

Proof: $E(x-m)^2=\sum_{i=1}^{n}(x-m)^2p_i$

$=\sum(x^2-2xm+m^2)p_i$

$=\sum(x^2)p_i- 2m\sum xp_i+m^2\sum p_i$

$=E(x^2)-2mE(x)+m^2$

$=E(x^2)-2m.m+m^2$

$=E(x^2)-m^2$

$=E(x^2)-(E(x))^2$

$E(x-E(x))^2$ is called the variance of the variable x and it is denoted by Var(x).

Corollary 2: If there are n random variables $x_1,x_2,x_3,\ldots,x_n$ each having mean $m_1, m_2, m_3,\ldots,m_n$.

$$Var(x_1+x_2+x_3+\ldots+x_n)=\sum_{i=1}^{n}Var(x_i)+2\sum_{i=i}^{n}i\neq jCov(x_i,x_j)$$

$Cov(x_i,x_j)=E(x_i-m_i)(x_j-m_j)$ when $\neq j$.

$$Var(x_1+x_2+x_3+\ldots+x_n)=E[(x_1+x_2+x_3+\ldots+x_n)-(m_1+m_2+m_3+\ldots+m_n)]^2$$

$$=\sum_{i=1}^{n}E(x_i-m_i)+2\sum_{i=1}^{n}i\neq j\,E(x_i-m_i)(x_j-m_j)$$

The first term in the above equation is Variance of the variable x_i whereas the second term is the $Cov(x_i,x_j)$. Therefore,

$$Var(x_1+x_2+x_3+\ldots+x_n)=\sum_{i=1}^{n}Var(x_i)+2\sum_{i=1}^{n}i\neq jCov(x_i,x_j)$$

If the variables are mutually independent then the covariance term in the above expression is zero (since mutual independence rules out joint occurrence).

Check Your Progress 5

1) A man purchases a lottery ticket. He may win first prize of Rs.10,000 with probability .0001 or the second prize of Rs. 4,000 with probability .0004. On an average, how much he can expect from the lottery?

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2) A box contains 4 white balls and 6 black balls. If 3 balls are drawn at random, find the mathematical expectation of the number of white balls.

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3) If y=a + b.x, where a and b are constants, prove that E(y)=a +bE(x).

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25.9 LET US SUM UP

The term probability in a very crude sense implies the chance factor and it issued frequently where there is uncertainty about something. Mathematicians from the very early ages tried endlessly to build up a structure under which this uncertain thing called probability could be analysed. In this unit we learnt about some basic terminology which is used in probability related matters. We discussed various methods frequently used to determine the probability of various non-deterministic experiments. Using these techniques, you can determine the probabilities of many day-to-day uncertain events. Further we learnt about the theorems of probability (total and compound) and conditional probability. Mathematical expectation introduced at the end of this unit is nothing but the mean of a random variable. It is quite useful in understanding the nature of a random variable.

25.10 KEY WORDS

Bayes' Theorem : If an event A occurs in conjunction with one of them mutually exclusive and collectively exhaustive events E_1, E_2, E_3….. E_n and if A actually occurs, then the probability that it was preceded by a particular event E_i(I=1,2,3……n)is given by

$$P(B_i \mid A) = P(A \mid B_i) \times P(A) / \sum_{i=1}^{n} P(B_i) P(A \mid B_i)$$

Conditional Events : If two events A and B are mutually exclusive and if it is known that event b has already taken place, the probability of A is known as the conditional probability of A given B. Symbolically P(A|B) = P (A ∩ B) / P(B).

Events : When some of the elements of the sample space of a random experiment satisfy a particular criterion, we call it an event.

Independent Events : Events are said to be independent of each other if occurrence of one event is not affected by the occurrence of the others. If events A and B are independent then P(A∩B)=P(A)×P(B).

Marginal Probability : In a bivariate distribution the probability that X will assume a given value X whatever the value of Y is called the marginal probability of Y. Similarly, one c and e fine the marginal distribution of Y.

Mathematical Expectation : If the variable x can take values $x_1, x_2, x_3,........., x_n$ with probabilities $p_1, p_2, p_3,........., p_n,$ then the mathematical expectation or the expected value of the variable, is defined as the weighted sum of the values of the variable the weights being the probabilities of the values of the variable. Expected value of a variable 'x' is denoted by E(x).

$$E(x) = x_i p_1 + x_2 p_2 + x_3 p_3 + ... + x_n p_n = \sum_{i=1}^{n} x_i p_i$$

Provided that $\sum_{i=1}^{n} p_i = 1$.

Mutually Exclusive Events : Events are said to be mutually exclusive when no two or more of them can occur simultaneously.

Mutually Exhaustive Events : Events are exhaustive if at least one of them necessarily occurs.

Sample Space : The collection of all possible outcomes of an experiment is called the sample space.

25.11 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) If the three children were born on three different days, the first child may be born in any one of 365 days, the second children has to be born on anyone of the remaining 364 days and the third child has to be born on anyone of the remaining 363 days. Therefore, the probability that the three children will be born on 3 different days in a year is 365.364.363/365.365.365.

2) Five persons can sit in a row in 5! = 5.4.3.2.1 = 120 ways. Considering a and b together they can arrange among themselves in 4! 2!= 48 ways. Therefore, the required probability is = 48/120 = .4.

3) Two cards can be drawn from a pack of 52 cards in $^{52}C_2$= 1326 ways. These outcomes are mutually exclusive, exhaustive and equally likely.

   a) The number of cases favourable to both the cards are red is $^{26}C_2$= 325. Therefore, that probability of drawing both red cards = 325/1326.

   b) One heart and one diamond can be drawn in 13.13= 169 ways. Therefore, probability of drawing one heart and one diamond= 169/1326.

4) One white ball could be drawnin$^{6}C_1$=6 ways and one ball can be drawn in $^{10}C_1$ = 10 ways. Therefore, the probability of drawing one white ball =6/10=.6.

5) The total number of ways of distributing n identical objects into r compartments is given by the formula $^{n+r-1}C_{r-1}$. Using the formula we can find out that there are 816 mutually exclusive exhaustive and equally likely ways of distributing 15 identical objects into 4 numbered boxes.

   a) If each box is to contain at least 2 objects, we place 2 objects in each box and then the remaining 7 objects could be distributed among the 4 boxes. Using the above formula, we get there are 120 ways of doing that. Therefore, the required probability is 5/34.

   b) If the number box has to be empty then it means there should be at least one object in each box. We first distribute 1 object in each box. Then the remaining 11 objects could be distributed in 364 ways (using the above-mentioned formula). Therefore, the required probability is given by 91/204.

6) Total number of possible equally likely exhaustive and exclusive ways of choosing 3 tickets is given by $^{20}C_3$ = 1140. The three numbers will be in A.P if the difference between the numbers is either 1 or 2 or 3 ……..9. If the

difference is 1 then 18 sets of numbers are possible, (123,234,345,………181920). Similarly, if the difference is 2 then 16 sets of numbers are possible, (135,246,357,………161820).Thus, we can find the total number of sets of 3numbers all of them being in A. P is 90. and the required probability is 3/38.

Check Your Progress 2

1) $P(A^c)=½, P(AUB)=7/12, P(A|B)=3/4, P(A^c \cap B)=1/12, P(A^c \cap B^c) = 5/12$, $P(A^c U B) = 3/4$.

2) Do yourself using Sub-section 25.5.1.1.

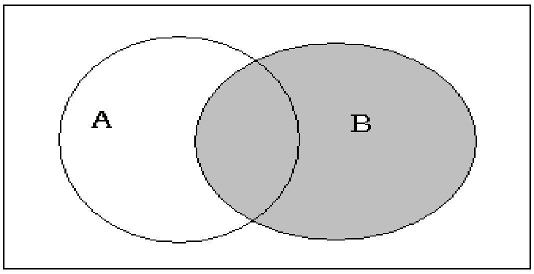
Check Your Progress 3

1) A: The sales person makes sale to the first customer.

B: The sales person makes sale to the second customer.

$P(A)= P(B)= ½$

(AU B): the sales person will make a sell.

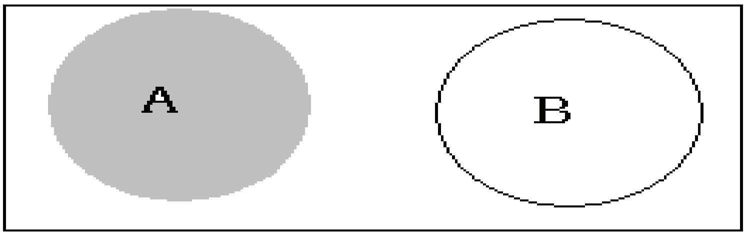


$P(AUB)=1-P(A^c \cap B^c)=1-P(A^c)(B^c)$[as events A and B are independent]

$=1–½. ½=¾$

2) Do yourself using Sub-section 25.6.2.

3)



$P(A \mid A \cup B) = P(A \cap (A \cup B)) / P(A) + P(B).[P(A \cup B) = P(A) + P(B)$ theorem of total probability]

$= P((A \cap A) \cup (A \cap B)) = P(A \cup (A \cap B)) = P(A) + P(A \cap B) - P(A \cap (A \cap B))$

$= P(A) + 0 - 0 = P(A)$

Therefore $P(A \mid A \cup B) = P(A) / P(A) + P(B)$

4) Hints: Let A denote the event of getting 3 white balls in the first draw and B denote the events of getting 3 red balls in the second draw. We need to find the compound probability P(AB) = P(A) P(B|A)

Check Your Progress 4

1) A: Selected bulb is defective.

B_1: Selected bulb has been produced by machine 'a'

B_2: Selected bulb has been produced by machine 'b'

B_3: Selected bulb has been produced by machine 'c'

Bi	P(Bi)	P(A|Bi)	P(Bi).P(A|Bi)
B1	0.25	0.05	0.0125
B2	0.35	0.04	0.014
B3	0.4	0.02	0.008
Total	1		0.0345

From Bayes' theorem

$$P(B_3 \mid A) = P(A \mid B_3) \times P(A) / \sum_{i=1}^{n} P(B_i)\, P(A \mid B_i) = .008 / 0.345 = 16/69.$$

2) Follow the same method. Answer=3/7.

Check Your Progress 5

1) The calculations for mathematical expectation are provided below

X	P(x)	x.P(x)
0	0.9995	0
4000	0.0004	1.6
10000	0.0001	1
Total		2.6

E(x)= 2.6Rs.

2) Probability of drawing '0' white ball = ${}^4C_0\,{}^6C_3/{}^{10}C_3$ = 1/6

Probability of drawing '1' white ball = ${}^4C_1\,{}^6C_2/{}^{10}C_3$ = ½

Probability of drawing '2' white ball = ${}^4C_2\,{}^6C_1/{}^{10}C_3$ = 3/10

Probability of drawing '3'white ball=${}^4C_3\,{}^6C_0/{}^{10}C_3$= 1/30

Fill up the following blank table to get the required expectation

X	p(x)	x.P(x)
0		
1		
2		
3		
Total		

3) Let x takes the values $x_1, x_2, ..., x_n$ with probabilities $p_1, p_2, ...p_n$ respectively, where $\sum p_i = 1$. Therefore, y will take the values:
$y_1 = ax_1 + b, y_2 = ax_2 + b, ..., y_n = ax_n + b$

$$E(y) = y_1 p_1 + y_2 p_2 + y_n p_n = (ax_1 + b)p_1 + (ax_2 + b)p_2 + ... + (ax_n + b)p_n$$
$$E(y) = (ax_1 p_1 + ax_2 p_2 + ... + ax_n p_n) + (bp_1 + bp_2 + ... + bp_n)$$

$$E(y) = \sum_{i=1}^{n} ax_i p_i + \sum_{i=1}^{n} bp_i = a\sum_{i=1}^{n} x_i p_i + \sum_{i=1}^{n} bp_i$$

25.12 EXERCISES

Q1. There are 5 green and 7 red balls. Two balls are selected one by one without replacement. Find the probability that first is green and second is red.

Ans. P(G) X P(R) = (5/12) X (7/11) = 35/132

Q2. Two cards are drawn from the pack of 52 cards. Find the probability that both are diamond or both are kings.

Ans. Total number of ways = ${}^{52}C_2$

Both are diamonds = ${}^{13}C_2$

Both are Kings = ${}^{4}C_2$

P (both are diamonds or both are kings) = $({}^{13}C_2 + {}^{4}C_2) / {}^{52}C_2$

= 0.063

Q3. A tyre manufacturing company kept a record of the distance covered before a tyre needed to be replaced. The table shows the results of 1000 cases

Distance (in Km)	Less than 4000	4001 to 9000	9001 to 14000	More than 14000
Frequency	20	210	325	445

If a tyre is bought from this company, what is the probability that:

i) it has to be substituted before 4000 km is covered?

ii) it will last more than 9000 km?

iii) it has to be replaced between 4000 km and 14000 km covered by it?

Ans. i) Total number of trials = 1000.

The frequency of a tyre required to be replaced before covering 4000 km = 20

So, $P(E_1) = 20/1000 = 0.02$

ii) The frequency that tyre will last more than 9000 km = 325 + 445 = 770

So, $P(E_2) = 770/1000 = 0.77$

iii) The frequency that tyre requires replacement between 4000 km and 14000 km = 210 + 325 = 535.

So, $P(E_3) = 535/1000 = 0.535$

Q4. A coin is tossed three times, consider the following events.

P: 'No head appears',

Q: 'Exactly one head appears' and

R: 'At Least two heads appear'.

Check whether they form a set of mutually exclusive and exhaustive events.

Ans: The sample space of the experiment is:

S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT} and

P = {TTT},

Q = {HTT, THT, TTH},

R = {HHT, HTH, THH, HHH}

$P \cup Q \cup R$ = {TTT, HTT, THT, TTH, HHT, HTH, THH, HHH} = S

Therefore, P, Q and R are exhaustive events.

and

$P \cap Q = \varphi$,$P \cap R = \varphi$ and$Q \cap R = \varphi$

Therefore, the events are mutually exclusive.

Hence, P, Q and R form a set of mutually exclusive and exhaustive events.

Q5. If $P(A) = 7/13$, $P(B) = 9/13$ and $P(A \cap B) = 4/13$, evaluate $P(A|B)$.

Ans: $P(A|B) = P(A \cap B)/P(B) = (4/13)/(9/13) = 4/9$

UNIT 26 PROBABILITY DISTRIBUTIONS-I

Structure

26.0 OBJECTIVES

After going through this unit, you should be able to

- discuss the concept of random variable;
- estimate the probability density function;
- find the probability mass function;
- distinguish between probability mass function and probability density function; and
- identify moments generating functions.

26.1 INTRODUCTION

The previous unit has discussed the theory of probability. This unit will introduce the concepts related to probability distribution. Random variables are always generated with a particular pattern of probability attached to them. Thus, based on the pattern of probability for the different values of random variable, we can distinguish them. Once we know these probability distributions and their properties, and if any random variable fits into a probability distribution, it will be possible to answer any question regarding the variable. In this unit, we have defined the random variable and made a broad distinction of the probability distributions based on whether the random variable is continuous or discrete. Then we have discussed how the moments of a probability distribution describe the distribution completely; how the technique of moment generating function could be used to obtain the moments of any probability distribution.

26.2 ELEMENTARY CONCEPT OF RANDOM VARIABLE

When a random experiment is performed, we are often not interested in the detailed results, but only in the value of some numerical quantity determined by the experiment. It is natural to refer to a quantity whose value is determined by the result of a random experiment as a random quantity.

A variable is called random when its occurrence depends on the chance factor. In other words, there is a probability that the variable will take a particular value. When we toss a coin, say thrice, the number of heads we obtain is a random variable. Suppose in this case, X denotes the number of heads. Clearly, X can take four values, 0, 1, 2, and 3. More importantly, there is a probability attached with every value of the variable X. Therefore, X is a random variable. We can cite several examples of random variable: the number of red cards drawn when we draw 10 cards from a pack of 52 cards, the number we obtain when a die is rolled, number of accidents in a city and the number of printing mistakes in a page of a newspaper.

Random variables could be either continuous or discrete. A discrete random variable takes only distinct values. In the above examples all are discrete variables. If Y is a variable that denotes the number when we roll a die, it will always take integral values. Practically, it will take any value among {1,2, 3, 4, 5, 6} and cannot take the value 5.5 or 4.3. Therefore, Y is a discrete random variable.

Similarly, a random variable is continuous if it takes infinite number of values within a given interval or range. The life of an electrical gadget, duration of phone calls received by a telephone operator and the amount of annual rainfall in a particular district of India are examples of continuous random variables. A random variable always takes real values. Therefore, we can define a random variable as a real valued function defined over the points of a sample space (set of all possible outcomes of an experiment) with a probability measure.

26.3 PROBABILITY MASS FUNCTION

Probability distribution of a random variable is a statement specifying the set of its possible values together with the respective probabilities. When a random experiment is theoretically assumed to serve as a model, the probabilities could be represented as a function of the random variable. Let a discrete random variable X assume the values $x_1, x_2, x_3 \ldots x_n$ with probabilities $p_1, p_2, p_3, \ldots\ldots\ldots, p_n$, satisfying the conditions: $\sum_{i=1}^{n} p_i = 1, 0 \le p \le 1$. The specification of the values of x_i together with the probabilities p_i defines the probability distribution of the random variable X. It is called the discrete probability distribution of X.

Often, the probability that the discrete random variable X assumes the value x is represented by f (x), where, f(x) = P (X = x) = probability that X assumes a specific value x. The function f(x) is known as the probability mass function(p.m.f.).

The discrete random variable X can assume countably infinite number of values with the p.m.f. f(x).The p.m.f. f(x) must satisfy the following two conditions:

1) $f(x) \geq 0$

2) $\sum_{i=1}^{n} f(x_i) = 1$

Example: An unbiased coin is tossed until the first head is obtained. If the random variable X denotes the number of tails preceding the first head, then what is the probability distribution of X?

Here $S = \{H, TH, TTH, TTTH, ...\}$ $S = \{H, TH, TTH, TTTH, \ldots\ldots..\}$.

Sample point	H	TH	TTH	………………
X= No. of tails preceding a head	0	1	2	

The probability distribution of the random variable X is shown in the table. Clearly, the p.m.f. of X is $F(X = x) = (1/2)^{n+1}$. It satisfies the two conditions mentioned earlier, i.e., f(x) is always greater than '0' and $\sum_{i=1}^{n} f(x_i) = 1$. Note that X is a countably infinite random variable.

Table 26.1: Probability Distribution of X

Values of X	f (X=x)
0	(½)
1	$(½)^2$
2	$(½)^3$
3	$(½)^4$
4	$(½)^5$
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N	$(½)^n$

$$= \sum_{i=1}^{n} f(x_i) = 1/2 + (1/2)^2 + (1/2)^3 + (1/2)^4 + ...(1/2)^n$$

$$=½\{1+ (½)+(½)^2+(½)^3\ldots\ldots\ldots+(½)^{n-1}\}$$

$$=½.1/(1-½) = \frac{1/2}{1/2}$$

$$=1$$

Example: Let X be a discrete random variable. Its probability distribution is given by Table 26.2. Its density could be represented by Fig. 26.1.

Table 26.2: Probability distribution of X

Values of X	f (X=x)
-3	0.216
-1	0.432
1	0.288
3	0.064

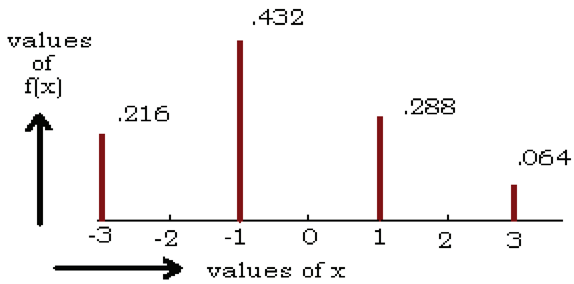

Fig. 26.1: Probability density of X

26.4 PROBABILITY DENSITY FUNCTION

In the previous section, we discussed the probability distribution of a discrete random variable. But what if the random variable is continuous, i.e., it can take any value within a given range. Since the number of possible values, a continuous variable can take is uncountably infinite, we cannot assign a probability to each value taken by the variable, as we did in case of the discrete random variable. In case of continuous random variable, we assign probability to an interval, which is in the range of the relevant random variable. Therefore, a continuous random variable has its probability density function as f(x) if it is a continuous non-negative function and it gives the probability that the random variable will lie in a specified interval when integrated over the interval. Let x be a continuous random variable which can assume any value in the interval (a, b) i.e., $a \leq x \leq b$.

Let $f(x)$f(x) be a non-negative function such that:

$$P(c \leq x \leq d) = \int_c^d f(x)dx$$

i.e., the probability of x lying in any given interval (c, d) can be obtained by integrating the function between the two limits of the interval.

The function f(x) is called the probability density function (p.d.f.) provided it satisfies the following two conditions

1) $f(x) \geq 0$

2) If the range of the continuous random variable is (a,b) such that $\int_a^b f(x) = 1$;

$\int_a^b f(x)dx = 1$where, $a \le x \le b$ is the range of x.

$P(c \le x \le d) =$ Area under the probability curve between the vertical line c and d (refer Fig. 26.2).

The shaded area in Fig. 26.2 represents the probability that the variable x will take values between the interval (c, d), whereas its range is (a,b). In the figure, we have taken the values of x in the horizontal axis and those of f(x) on the vertical axis. This is known as the probability curve. Since f(x) is a p.d.f., total area under the probability curve is 1 (i.e., $\int_a^b f(x) = 1)\int_a^b f(x)dx =$ 1) and the curve cannot lie below the horizontal axis as f(x) cannot take negative values. Any function that satisfies the above two conditions, can serve as a probability density function in the presence of real values only.

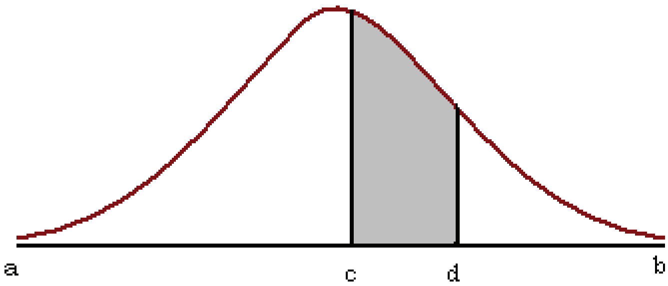

Fig. 26.2: Probability curve

Example: If x is a continuous random variable with the probability density function f(x)=k.e^{-3x}for x>0 and 0otherwise, then find k and P(0.5 ≤ x≤ 1)

If the function specified in the problem is to serve as ap.d.f., then the following two conditions must hold:

1) f(x) ≥0 for all values of x

2) If the range of the continuous random variable is (a,b) $\int_a^b f(x) = 1\int_a^b f(x)dx = 1$

Clearly, the range of the function given in the problem is -∝ to ∝ and for every value of x within that interval, f(x) is positive, provided that 'k' is positive. To satisfy these condition.

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

or, $\int_0^{+\infty} f(x)dx + \int_{-\infty}^{0} f(x)dx = 1$

or, $\int_0^{+\infty} f(x)dx = 1\int_0^{+\infty} f(x)dx = 1$

or $\int_{0}^{+\infty} ke^{-3x}\,dx = 1 \int_0^{+\infty} ke$

or, k/3 = 1

or, k =3

Thus ,we get ,f(x)=3.e^{-3x} for x>0 and 0 otherwise.

$$P(0.5 \le X \le 1) = \int_{0.5}^{1} 3e^{-3x}dx = -e^{-3} + e^{-1.5} = 0.173$$

26.5 PROBABILITY DISTRIBUTION FUNCTION

The Cumulative distribution function (c.d.f) - F(x), represents the probability that the random variable takes a value less than or equal to a specified value x. Thus F(x) shows the area under probability curve f(x) to the left of the ordinate at any specified value. Thus, for example:

$F(c) = P(x \le c) = \int_{-\infty}^{c} f(x)dx$ i.e., Area under the probability curve to the left of the vertical line c.

$P(c \le x \le d) = P(c \le x \le d)$ = Area between the ordinates c and d = (Area to the left of the ordinate at d minus (Area to the left of the ordinate at c) = F(c) – F(d)

Thus, $P(c \le x \le c) = F(c) - F(c)\, P(c \le x \le c)$ = Area between the ordinates c and c, which is 0. i.e., $P(x = c) = 0$. Thus, for continuous probability distribution a random variable takes any specific value is zero.

The distribution function of a continuous random variable has the same nice properties as that of a discrete random variable viz.,

i) F(-∞) = 0, F(∞) = 1

ii) If a<b, then F(a) ≤ F(b) where and bare any real number.

iii) Furthermore, it follows directly from the definition that P (a ≤ x ≤ b) = F(b) – F(a), where a and b are real constants with a ≤ b

iv) F(x) = d/dx F(x) where the derivatives exist

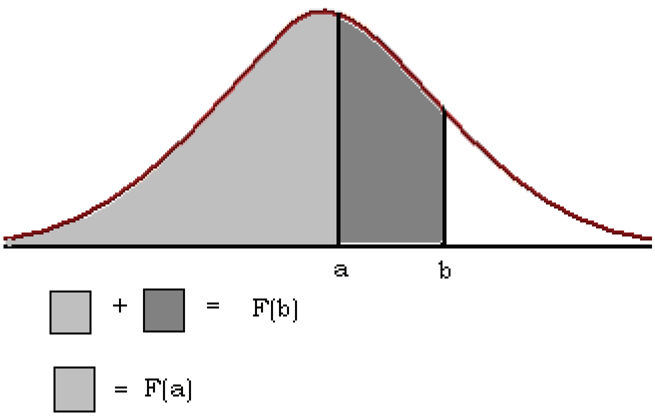

Fig. 26.3: Distribution of a Continuous Random Variable

Example: Find the distribution function of the random variable x of the previous example and use it to re-evaluate $P(0.5 \le x \le 1)$

For all non-positive values of x, f(x) takes the value 0. Therefore,

F(x)=0forx≤ 0.

For x>0,

$$F(x) = \int_0^x 3e^{-3t}\,dt + \int_x^\infty f(t)\,dt$$

Thus, F(x)=0forx≤0,andF(x)=1–e^{-3x}forx>0.

To determine $P(0.5 \le x \le 1)$,we use the fourth property of the distribution function of a continuous random variable.

P(0.5) ≤x)≤1)=F(1)- F(0.5) =(1–$e^{-3)}$–(1–$e^{-3\times 0.5}$)=0.173

Check Your Progress 1

1) If X is a discrete random variable having the following p.m.f.,

X	0	1	2	3	4	5	6	7
P(X = x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

   i) determine the value of the constant k
   ii) find P(X<5)
   iii) find P(X>5)

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2) For each of the following, determine whether the given function can serve as a p.m.f.

   i) f(x) = (x - 2)/5 for x = 1,2,3,4,5
   ii) f(x) = x^2/30 for x = 0,1,2,3,4
   iii) f(x)= 1/5 for x=0,1,2,3,4,5

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26.6 MOMENTS AND MOMENTS GENERATING FUNCTIONS

26.6.1 Moments

Let X be a discrete random variable, which takes the values $x_1, x_2, x_3, \ldots\ldots, x_n$ and f(x) is the probability that X will take the value x. Then the expected value of X is given by

$$E(X) = \sum_{i=1}^{n} x_i f(x_i)$$

Correspondingly, if X is a continuous random variable and f(x)gives the probability density at x, the expected value of x is given by $E(X) = \int_{-\infty}^{\infty} x f(x)\,dx$

In the definition of continuous random variable, we assume that the integral exists, otherwise, the mathematical expectation is undefined. The expected value of a random variable gives its mean value if we think f(x) as the relative frequency of the random variable X when it takes the value x.

Again, if g(x) is a continuous function of the value of the random variable X, then the expected value of g(x) is given by $E(g(x)) = \sum_{i=1}^{n} g(x_i)\, f(x_i)$, when X is a discrete random variable. When it is a continuous random variable, the expected value of g(x) is given by $E(g(x)) = \int_{-\infty}^{\infty} g(x) f(x)\,dx$.

The determination of mathematical expectation can often be simplified by using the following theorems:

1) E(a+ b.X)= a +b.E(X)
2) E(a)= a, where a is a constant.
3) If $c_1, c_2, c_3, \ldots\ldots, c_n$ are constants, $E\left[\sum_{i=1}^{n} c_i g(x_i)\right] = \sum_{i=1}^{n} c_i E[g(x_i)]$

In statistics as well as economics, the notion of mathematical expectation is very important. It is a special kind of moment. We now introduce the concept of moments and moment generating functions.

The r^{th} **order moment about the origin** of a random variable x is denoted by $\boldsymbol{\mu_r'}$ and is given by the expected value of x^r.

Symbolically, for x being a discrete random variable $\mu_r' = E(x^r) = \Sigma_x x^r f(x) \mu_r' =$ $E(x^r)$ for r = 1, 2,3,….,n

For x being a continuous random variable $\mu_r' = E(x^r) = \int_{-\infty}^{\infty} x^r f(x)\,dx$

The term moment comes from physics. If f(x) symbolizes quantities of points of masses, where x is discrete, acting perpendicularly on the x axis at distance x from the origin, then $\mu_1' \mu_1'$ as defined earlier would give the center of gravity, which is the first moment about the origin. Similarly, μ_2' gives the moments of inertia.

When $r = 0$, we have $\mu_0' = E(x^0) = E(1) = \int_{-\infty}^{+\infty} f(x)dx = 1$. When $r = 1$, we have $\mu_1' = E(x^1) = E(x)$. In statistics, μ_1' gives the mean of a random variable and it is generally denoted by μ.

The special moments we shall define are of importance in statistics because they are useful in defining the shape of the distribution of a random variable,viz., the shape of it's probability distribution or it's probability density.

The **rth moment** of random variable about mean is given by μ_r. It is the expected value of $(X-\mu)^r$, symbolically $\mu^r = E((X-\mu)^r) = \sum_{i=1}^{n}(x_i-\mu)^r f(x_i)$, for r=1,2,3,…..,n, when x is discrete and om variable and when x is a continuous random variable, rth moment is given by $\mu_r = \int_{-\infty}^{\infty}(x-\mu)^r . f(x)dx$.

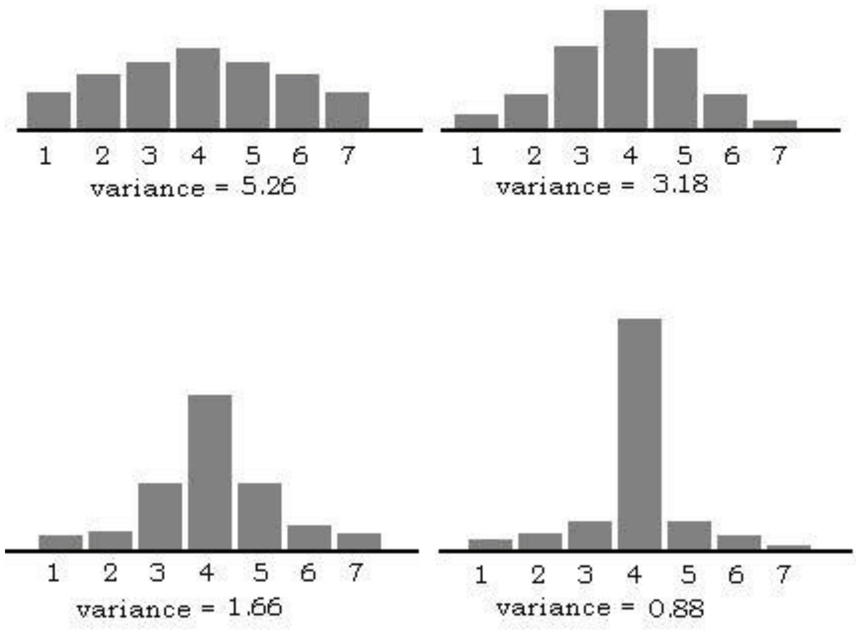

Fig. 26.4: Distributions with same Means but different Variances

Note that, $\mu_0 = 1$ and $\mu_1 = 0$ for any random variable for which μ exists. The second moment about mean i.e., $\mu_2 = E(x-\mu)^2$ is called the variance and denoted by σ^2 or $Var(x)$. σ^2orVar(x)

The above example shows how probability distributions vary with the variance of the random variable. A high value of σ^2 means the spread of the distribution is thick at the tails and a low value of σ^2 implies the spread of the distribution is tall at the mean of the distribution and the tails are flat. Similarly, third order moment about mean describes the symmetry or skewness (i.e., lack of symmetry) of the distribution of a random variable.

We state a few important theorems on moments without going into details as they have been covered in the unit on probability.

1) $\sigma^2 = \mu_2' - \mu^2 \sigma^2$

2) If the random variable X has the variance σ^2, then $Var(aX+b) = a^2\sigma^2$

3) **Chebyshev's theorem**: To determine how σ or σ^2 is indicative of the spread or the dispersion of the distribution of the random variable, Chebyshev's theorem is very useful. Here we will only state the theorem:

If μ and σ^2 are the mean and variance of a random variable, say X, then for any constant k

$P(|X-\mu|<k.\sigma)\geq 1-1/k^2 \;;\; \sigma \neq 0$

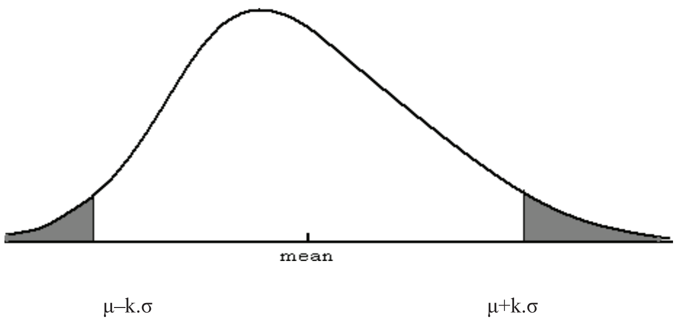

Fig 26.5: Chebyshev's Theorem

26.6.2 Moment Generating Functions

Although moments of most distributions can be determined directly by evaluating the necessary integrals or sums, there is an alternative procedure, which some times provides consider able simplifications. This technique is known as the technique of moment generating function.

The moment generating function of a random variable X, where it exists is given by,

$M_X(t) = E(e^{tX}) = \sum_{i=1}^{n} e^{tx} f(x);$ when X is discrete and

$M_X(t) = E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} f(x)\,dx$; when X is continuous

The independent variable is t, and we are interested in values of t in the neighbourhood of 0. To explain why, we refer to this function as a "moment generating function". Let us substitute for e^{tx} with its Maclaurin's series expansion as follows:

$e^{tx}=1+tx+t^2x^2/2\,!+t^3x^3/3\,!+\ldots\ldots\ldots+t^rx^r/r\,!+\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots..$

Thus, for discrete case we get,

$M_x(t)=E(e^{tX})= \sum_{i=1}^{n} e^{tx} f(x) = \sum_{i=1}^{n} f(x) \sum_{i=1}^{n} e^{tx} f$

$(1+tx+t^2x^2/2\,!+t^3x^3/3\,!+\dots\dots+t^rx^r/r\,!+\dots\dots)$

$$= \sum_{i=1}^{n} f(x) + t\sum_{i=1}^{n} xf(x) + t^2/2! \sum_{i=1}^{n} x^2 f(x) + t^3/3! \sum_{i=1}^{n} x^3 f(x) + \dots + t^r/r! \sum_{i=1}^{n} x^r f(x) + \dots$$

$= 1 + t.\mu + t^2/2\,!.\ M'_2 + \dots + t^r/r\,!.\ \mu'_r + \dots$

Thus, we can see that in the Maclaurin's series of the moment generating function of X, the coefficient of $t^r/r!$ is μ'_r, which is nothing but the r^{th} order moment about the origin. In the continuous case, the argument is the same.

To get the r^{th} order moment about the origin, we differentiate $M_x(t)$ r times with respect to t and put t=0 in the expression obtained. Symbolically,

$\mu'_r = d^rM_x(t)/dt^r|_{t=0}$

An example will provide more clarity.

Example: Find the moment generating function of the random variable whose probability density is given by

F(x)=e^{-x} if x>0

0 otherwise

and use that to find the expression for μ'_r.

By definition,

$$M_x(t) = E(e^{tX}) = \int_{-\infty}^{+\infty} e^{tr} f(x)dx = \int_{0}^{\infty} e^{tx} e^{-x}\,dx = \int_{0}^{\infty} e^{-x(1-t)}dx = 1/1-t;\ \text{for}\ t<1$$

When |t|<1,the Maclaurin's series for this moment generating function is

$M_x(t) = 1+t+t^2+t^3+t^4+\dots+t^r+\dots=1+ 1\,!.t/1\,!+2\,!.t^2/2\,! +3\,!.t^3$
$/3\,!+\dots+r\,!.t^r/r\,!+\dots$

Hence, $\mu'_r = d^rM_x(t)/dt^r|_{t=0} = r\,!$ for r = 0,1,2…

If 'a' and 'b' are constants, then

1) $M_{x+a}(t) = E(e^{t(x+a)}) = e^{at}.M_x(t)$

2) $M_{x.b}(t) = E(e^{t.b.x}) = M_x(b.t)$

3) $M(x+a)/b(t) = E(e^{t(x+a)/b}) = e^{(a/b)t}.M_x(t/b)$

Among the above three results, the third one is of special importance. When $a = -\mu$ and $b = \sigma, M_{(x-\mu)/\sigma}(t) = E(e^{t(x-\mu)/\sigma}) = e^{(-\mu/\sigma)t}.M_x(t/\sigma)$

Check Your Progress 2

26) Given X has the probability distribution $f(x)=1/8.{}^3C_x$ for x=0,1,2, and 3, find the moment generating function of this random variable and use it to determine μ_1' and μ_2'.

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26.7 LET US SUM UP

In this unit, we learnt about the various concepts related to probability distribution. We understood how random variables are always generated with a particular pattern of probability attached to them. Based on that, we discussed the probability distributions and their properties. We defined the random variable and made a broad distinction of the probability distributions based on whether the random variable is continuous or discrete. Then we discussed how the moments of a probability distribution describes the distribution completely; how the technique of moment generating function could be used to obtain the moments of any probability distribution.

26.8 KEY WORDS

Continuous Random Variable : If a random variable can take any value with in its range, then it is called a continuous random variable.

Discrete Random Variable : If a random variable takes only countable number of values and there are no possible values of the variable located between two juxt a posed values of the variable, then it is called a discrete random variable.

Probability Distribution : Probability distribution of a random variable is a statement specifying the set of its possible values together with the respective probabilities

Probability Mass Function : Often the probability of the discrete random variable X assuming the value x is represented by f(x), f(x) = P (X = x) = probability that X assumes value x. The function f(x) is known as the probability mass function (p.m.f.) given that it satisfies following two conditions:

1) $f(x) \geq 0$

2) $\sum_{i=1}^{n} f(x_i) = 1 \sum_{i=1}^{n} f(x$

Probability Density Function : A continuous random variable having a probability distribution f(x) which is a continuous on-negative function, gives the probability that the random variable will lie in a specified interval when integrated over the interval, i.e.,

$$P(c \leq x \leq d) = \int_{c}^{d} f(x)dx$$

The function f(x) is called the probability density function (p.d.f.) provided it satisfies following two conditions

1) $f(x) \geq 0$
2) If the range of the continuous random variable is (a,b) integration $\int_{a}^{b} f(x)dx = 1$

26.9 ANSWERS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) A p.m.f. must satisfy the following two properties

   i) $f(x) \geq 0$

   ii) $\sum_{i=1}^{n} f(x_i) = 1 \sum_{1}^{n} f(xi) = 1$

   Using the second property, we get $10k^2 + 9k - 1 = 0$. It gives two values of k, viz., -1 and 1/10. Clearly, k cannot take the value -1 (as f(X = 1) = k and f(x)is always non-negative).Given k =1/10 rest is merely algebra.

2) i) Cannot, (since, f(x=1) < 0 and probability cannot be negative

   ii) Yes, can be a p.m.f since $(f(x) \geq 0$ and $\sum_{0}^{4} f(x) = 1$

   iii) Cannot since $(f(x)) \geq 0$ but $\sum_{0}^{4} f(x) \neq 1$

Check Your Progress 2

1) $M_x(t) = E(e^{tx}) = \sum_{i=0}^{3} e^{tx} \frac{1}{8} {}^3C_x$ = 1/8 (1+ $3e^t + 3e2^t + e3^t$) = 1/8 (1 + e^t)

   $\mu_1 = M_x(0) = 3/2$

   $\mu_2 = M_x(0) = 3$

26.10 EXERCISES

Q1. A fair coin is tossed twice. Let X be defined as the no. of heads shown. Find the range of X and its probability mass function P_X.

Ans. The sample space S = {HH, TT, HT, TH}

The no. of heads can be 0,1,or 2.

So, R(x) = {0,1,2}

The probability mass function $P_X(x) = P(X = x)$ for x = 0,1,2

$P_X(0) = P(X = 0) = 1/4$ [i.e. probability of getting no head]

$P_X(1) = P(X = 1) = 1/4 + 1/4 = 1/2$

$P_X(2) = P(X= 2) = 1/4$

Q2. From the below given p.m.f, find the value of r.

Y	0	1	2	3
P(Y=y)	0	2r	3r	$6r^2$

Ans. We know, $\sum P(y_i)=1$.

Thus,

$0 + 2r + 3r + 6r^2 = 1$

$5r + 6r^2 - 1 = 0$

$6r^2 + 6r – r -1 = 0$

$6r(r + 1) -1(r + 1) = 0$

$(6r – 1) (r + 1) = 0$

So, $6r – 1 = 0$ and $r + 1 = 0$

$r = -1$ is not possible as the value of probability lies between 0 and 1.

Thus, r = 1/6

Therefore, the value of r from the above stated PMF is 1/6.

Q3 What are the two conditions for probability density function?

Ans. The probability density function will only be valid if it meets the following conditions:

1) f(x) should be non-negative for all values of the random variable.
2) The area underneath f(x) should be equal to 1.

Q4. What is the difference between the probability mass function and the probability density function?

Ans. The main difference between the probability mass function (PMF) and the probability density function (PDF) is that PDF is used to describe the continuous probability distribution whereas PMF is used to describe the discrete probability distribution.

UNIT 27 PROBABILITY DISTRIBUTIONS-II

Structure

27.0 OBJECTIVES

After going through this unit, you should be able to

- discuss the concept of joint and marginal distribution;
- explain the three most common probability distributions (binomial, poisson and normal); and
- solve questions related to these three distributions.

27.1 INTRODUCTION

The previous unit has elaborated on the concept of random variables and the basics of probability distribution. This understanding would serve as the basis of the topics that we are going to cover in this unit. Uptil now, we have introduced the probability models involving only a single random variable. This unit will explain how probability modelling changes when we are interested in more than one random variable at the same time. Therefore, we would discuss about joint distribution, the associated marginal distribution and further the case of how to represent these when variables are independent. Additionally, three very commonly used probability distributions would be discussed, namely: binomial distribution, poisson distribution and normal distribution.

27.2 JOINT DISTRIBUTION

After having discussed about the random variables and the basics of probability distribution, we are now proceeding to the topic of joint distribution.

Very commonly, there might arise situations where we are dealing with more than one random variable. This concept of joint distribution allows us to understand the relationship between multiple events happening at the same time, e.g., from a deck of 52 cards, the joint probability of picking up a card that is both red and 6

Let us denote two events A and B and these be described by random variables X at value x and Y at value y. So, P(A ∩ B) = P(X = x ∩ Y = y) = P(X = x, Y = y) and this expression is referred to as the joint probability of X=x and Y=y.

More formal way to describe the same concept is as follows:

If X and Y are discrete random variables, then the function given by f(x,y)= P(X=x, Y=y) for each pair of values (x,y) within the range of X is called the joint probability distribution of X and Y.

Thus a bivariate fucntion can serve as the **joint proability distribution** of a pair of discrete random variables X and Y if and only if the values of f(x, y) satisfies the condition:

i) $f(x,y) \geq 0$ for each pair of values (x,y) within its domain

ii) $\sum_x \sum_y f(x,y)$ where the double summation extends over all possible points (x,y) within its domain.

Example: Determine the value of k for which the fucntion given by $f(x,y) = kxy;$ for x = 1, 2, 3; y = 1, 2, 3 can serve as a joint distrbution.

Ans. Substituting the values of x and y we get: $f(x,=1,y=1)=k; f(1,2)=2k; f(1,3)=3k;$
$f(2,1)=2k; f(2,2)=4k, f(2,3)=6k; f(3,1)=3k;$ and $f(3,2)=6k, f(3,3)=9k$

We now have to check the condtions for the joint probability distribution: k > 0 and

$$\sum_x \sum_y f(x,y) = k+2k+3k+2k+4k+6k+3k+6k+9k=1$$

So that 36k = 1 and $k = \frac{1}{36}$.

If we have more than two discrete random variables i.e., $X_1, X_2, ..., X_n,$ then the joint p.m.f of the variables is the function,

$$p(x_1, x_2, ..., x_n) = P(X_1 = x_1, X_2 = x_2, ..., X_n = x_n)$$

If the variables are continuous, the joint p.d.f of $X_1, ..., X_n$ is the function $f(x_1, x_2, ..., x_n)$ such that for any n intervals $[a_1, b_1], ..., [a_n, b_n],$

$$P(a_1 \le X_1 \le b_1, ..., a_n \le X_n \le b_n) = \int_{a_1}^{b_1} ... \int_{a_n}^{b_n} f(x_1, ..., x_n) dx_n ... dx_1$$

Example: Assume we have a corpus of 100 words. Table 27.1 depicts the words and the associated frequencies and probabilities of each of those words.

Table 27.1: Frequencies and Probabilities of 100 words

W	c(w)	P(w)	X	y
Shall	5	0.05	5	1
Up	8	0.08	2	1
Next	5	0.05	4	1
As	5	0.05	2	1
Fore	3	0.03	4	2
To	10	0.1	2	1
We	12	0.12	2	1
Some	8	0.08	4	2
Will	30	0.3	4	1
BBC	14	0.14	3	0

Here random variable X denotes the length of the word and Y denotes the number of vowels in the word.

Now, let us calculate f(x,y)= P(X=x, Y=y) taking different values of the random variable defined here (refer Table 27.2).

$f(x = 4, y = 2) = P(\text{Fore}) + P(\text{Some}) = 0.03+0.08 = 0.11$

$f(x = 3, y = 0) = P(\text{BBC}) = 0.14$

Table 27.2: Calculation of f(x, y)

		x			
		2	3	4	5
y	0	0	0.14	0	0
	1	0.35	0.3	0.05	0.05
	2	0	0	0.11	0

Moving forward, it becomes imperative to also mention about the special case of the theorem of joint distribution of independent random variables. According to this theorem, if X and Y are independent, then $f(x,y) = f_X(x).f_Y(y)$ for all values of x and y. This can also be stated in an alternative manner: We say that the random variables $X_1, X_2, . . ., X_n$ are independent if for every subset $X_{i_1}, X_{i_2}, ..., X_{i_k}$ of the variables, the joint p.m.f or p.d.f of the subset is equal to the product of the marginal p.m.fs or p.d.fs.

If X and Y are discrete random variables the function given by

$$F(x,y) = P(X \le x, Y \le y) = \sum_{x \le s} \sum_{y \le t} f(s,t); \text{ for } -\infty < x < \infty; -\infty < y < \infty$$

where, f(s,t) is the value of the joint probability distribution of X and Y at (s, t) is called the joint distribution function or the joint cumulative distribution of X and Y.

A bivariate function with values f(x, y) defined over xy plane is called **joint probability density function** of two continuous random variables X and Y if and only if

$P[X, Y \in A] = \int_A f(x,y)\,dxdy;$ for any region in the x y plane.

Similarly, a bivariate function can serve as joint probability density function of a pair of continuous random variable if its values f(x, y) satisfy the conditions

i) $f(x,y) \ge 0; -\infty < x < \infty; -\infty < y < \infty$

ii) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y)\,dxdy = 1$

Example: Given the joint probability density function of two random variable X and Y

$$f(x,y) = \begin{cases} x(x+y)\frac{3}{5}; \; for\, 0 < x < 1/2;\, 1 < y < 2 \\ 0; \; elsewhere \end{cases}$$

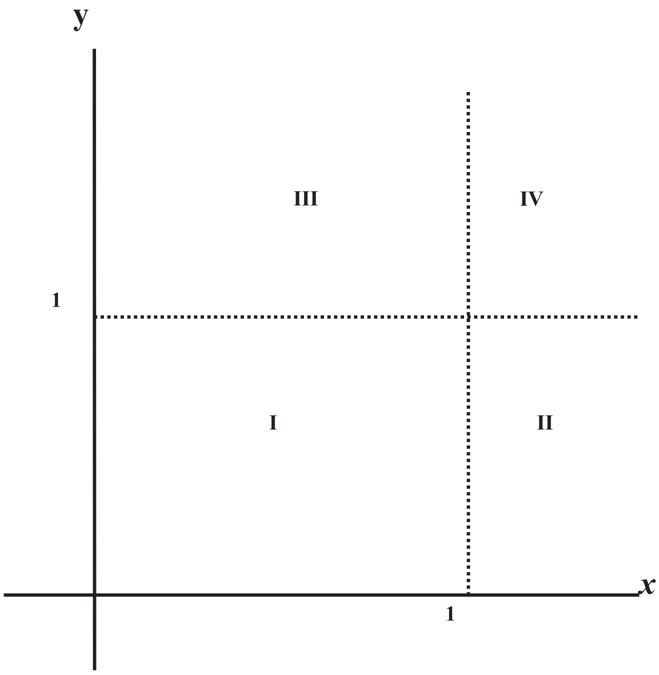

Fig. 27.1: Joint Probability Density Function of X and Y

Find $P[X,Y \in A]$ where A is the region $(x,y) \mid 0 < x < 1/2; 1 < y < 2$

$$P[X,Y \in A] = P(0 < x < 1/2; 1 < y < 2 \mid$$

$$= \int_1^2 \int_0^{1/2} \frac{3}{5} x(y+x)\,dxdy$$

$$= \int_1^2 \left[\frac{3yx^2}{10} + \frac{3x^3}{15}\right] dy = \int_1^2 \left(\frac{3y}{40} + \frac{1}{40}\right) dy = \left(\frac{3y^2}{80} + \frac{1}{40}\right)_1^2 = \frac{11}{80}$$

If X and Y are continuous random variables, the cumulative density or distribution function is given by

$$F(x,y) = P(X \le x, Y \le y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(s,t)\,ds\,dt;$$

for $-\infty < x < \infty; -\infty < y$

where, f(s,t) is the value of the joint probability distribution of X and Y at (s, t) and is called the joint distribution or the **joint distribution function of X and Y.**

We assume that the joint distribution is continuous everywhere and partially differentiable with respect to each variable, for all but a finite set of values of the two random variables and is analogous of the relationship $f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y);$ whenever this partial derivative exists. The joint distribution of two continuous variables determines their joint density at all point (x, y), where the joint density is continuous.

Example: $f(x,y) = \begin{cases} (x+y) \text{ for } 0 < x < 1; 0 < y < 1 \\ 0; \text{elsewhere} \end{cases}$

Region 1: For 0 < x < 1; 0 < y < 1; we get

$$F(x,y) = \int_0^y \int_0^x (x+y)\,dx\,dy = \frac{1}{2} xy(x+y);$$

Region II: For x > 1; 0 < y < 1, we get

$$F(x,y) = \int_0^y \int_0^1 (x+y)\,dx\,dy = \frac{1}{2} y(1+y)$$

Region III: For 0 < x < 1, y < 1; we get

$$F(x,y) = \int_0^1 \int_0^x (x+y)\,dx\,dy = \frac{1}{2} x(1+x)$$

Region IV: For x > 1, y > 1; we get

$$F(x,y) = \int_0^1 \int_0^1 (x+y)\,dx\,dy = 1$$

Since the joint distribution function is continuous everywhere, the boundaries between any two of these regions can be included in either one and we can write

$$F(x,y)=\begin{cases}0, & 0\le x\le 0, y\le 0\\ \frac{1}{2}xy(x+y); 0<x<1, 0<y<1\\ \frac{1}{2}y(y+1); x\ge 1, 0<y<1\\ x(x+1); 0<x<1, y\ge 1\\ 1; & x\ge 1, y\ge 1\end{cases}$$

27.3 MARGINAL DISTRIBUTION

Now, there would be cases where we might need to remove the influence of an event. This might happen if some events are irrelevant for some experiment or we could alternatively be only interested in a subset of the events. Therefore, we resort to 'marginalisation' which refers to the process of removing/omitting the influence of one or more events.

If X and Y are discrete random variables and f (x, y) is the value of their joint probability distribution at (x, y), then the function given by

$g(x)=\sum_{y} f(x,y)$ for each value of x within the range of X is called the **marginal distribution of X** and correspondingly the function given by $h(y)=\sum_{x} f(x,y)$ or each value of y within the range of Y is called the **marginal distribution of Y.**

If X and Y are the **continuous random variables** and f(x,y) is the value of their joint probability density at (x,y) then the function given by $g(x)=\int_{-\infty}^{\infty} f(x,y)\,dy; \text{for} -\infty<x<\infty$ is called the **marginal density of X** and the corresponding function given by $h(y)=\int_{-\infty}^{\infty} f(x,y)\,dx; \text{ for } -\infty<y<\infty$ is called the **marginal density of Y**

Example: We continue the example discussed for joint distribution and calculate marginal distribution (Refer Table 27.3& 27.4):

Table 27.3: Marginal Distribution of X

		X			
		2	3	4	5
y	0	0	0.14	0	0
	1	0.35	0.3	0.05	0.05
	2	0	0	0.11	0
	$\sum_y f(x,y)$	0.35	0.44	0.16	0.05

Table 27.4 Marginal Distribution of Y

		X				
		2	3	4	5	$\sum_x f(x,y)$
y	0	0	0.14	0	0	0.14
	1	0.35	0.3	0.05	0.05	0.75
	2	0	0	0.11	0	0.11

Check Your Progress 1

1) Let the joint probability density function for (X, Y) be

$$f(x,y) = \frac{x+y}{4}, x > 0, y > 0,\ 3x + y < 3,$$

zero otherwise.

Find the marginal probability density function of i) X, $f_X(x)$ ii) Y, $f_Y(y)$

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2) Given the joint probability density

$$f(x,y) = \begin{cases} \frac{2}{3}(x+2y); \text{for } 0 < x < 1, 0 < y < 1 \\ 0, \text{ elsewhere} \end{cases}$$

Find the marginal densities of X and Y

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27.4 THREE IMPORTANT PROBABILITY DISTRIBUTIONS

In real life, we come across various situations that can be modelled using probability distributions. This helps in understanding the probability of a certain occurrence. Therefore, based on distinct characteristics of different situations under consideration, we resort to the different probability distribution that best suits the characteristics.

The following section will deal with the three most common probability distributions. Here, we start off with the discussion of Binomial distribution leading to Poisson and then lastly Normal distribution.

27.4.1 Binomial Distribution

Repeated trials play has a very important role in probability and statistics, especially when the number of trials is fixed and the probability of success (or failure) in each of the trial is independent.

The theory, which we discuss in this section, has many applications; for example, it applies to events like the probability of getting 5 heads in 12 flips of a coin or the probability that 3 persons out of 10 having a tropical disease will recover. To apply binomial distribution in these cases, the probability (of getting head in each flip and recovering from the tropical disease for each person) should exactly be the same. More importantly, each of the coin tossed and chance of recovering of each patient should be independent.

Let x be a discrete random variable following a binomial distribution. The sampling distribution of x is the probability mass function (p.m.f) and is defined by $f(x) = {}_nC_x P^x Q^{n-x}; x = 0, 1, 2, \ldots, n; (P + Q) = 1;$ where P and Q are positive fractions.

Suppose that we have a series of independent trials, in each of which the probability of occurrence of an event is fixed and constant (say P). Let us denote the occurrence of the event as success and its complement that is the nonoccurrence of the event as failure. Then the probability that an event occurs exactly r times in n trails is ${}_nC_r P^r Q^{n-r};$ where Q = 1 – P and r may assume values = 0, 1, 2, …, n. Thus, in a series of n independent trials, if the probability of success in each trial is a constant (say P) and the probability of failure is Q, then the probability of x success (and obviously n – x failure) is given by the Binomial distribution. The two constants n and P appearing in the p.m.f f(x) are known as the parameters of Binomial distribution. If the values of n and P are known the distribution is completely known. It should be noted that the binomial distribution holds under following conditions:

1) The outcomes of any trial can be classified under only two categories (say success and failure)
2) The probability of success of each trials remain constant and does not change from one trial to another
3) The trials are independent, so that the probability of success in any trial remains unaffected by the results of other trials.

Example: Find the probability of getting 7 heads and 5 tails in 12 tosses of an unbiased coin.

Substituting x = 7, n = 12, p = ½ in the formula of binomial distribution, we get the desired probability of getting 7 heads and 5 tails.

$B(7;12,1/2)={}^{12}C_7.1/2^7.(1-1/2)^5={}^{12}C_7.(1/2)^{12}$

There are a few important properties of the binomial distribution. While discussing these, we retain the notations used earlier in this unit.

- Binomial distribution is a discrete probability distribution where the random variable assumes a finite number of values (0, 1, 2, 3, …,n). The distribution is specified by two parameters n and P.
- Mean = n P; Variance = nPQ (where n is the number of repeated trials and P and Q are the probability of success and failure).
- Binomial distribution may have either one or two modes. When (n+1)P is not an integer, mode is the largest integer contained therein. However, when (n+1)p is an integer, there are two modes given by (n + 1)P and {(n +1)P–1}.
- Skewness of binomial distribution is given by $=\frac{Q-P}{\sqrt{nPQ}}$; Kurtosis $=\frac{1-6PQ}{nPQ}$.

  When P = Q = ½ the distribution becomes symmetric for all values of n.
- If X and Y are two random variables, where X follows binomial distribution with parameters (n_1, p) and Y follows binomial distribution with parameters (n_2, p), then the random variable (X+Y) also follows binomial distribution with parameters (n_1+n_2,p).
- The moment generating function of a binomial distribution is given by

  $$M_x(t)=E(e^{tx})=\sum_{i=0}^{n}e^{tx}f(x)=\sum_{i=0}^{n}e^{tx}.{}^nC_x.p^x(1-p)^{n-x}$$

  $=\sum_{i=0}^{n}{}^nC_x(pe^t)^x(1-p)^{n-x}$. The summation is easily recognized as the binomial expansion of $\{p.e^t+(1-p)\}^n=\{1+p(e^t-1)\}^n$

We can derive the mean and variance of the binomial distribution using the moment generating function.

If we differentiate M_x (t) with respect to t twice, we get:

$$M'x(t)=npe^t[1+p(e^t-1)]^{n-1}$$

$$M''_x(t)=npe^t[(1+p(e^t-1)]^{n-1}+n(n-1)e^{2t}p^2[1+p(e^t-1)]^{n-2}$$

$$M''_x(t)=npe^t(1-p+npe^t)[1+p(e^t-1)]^{n-2}$$

And substituting t = 0, we get $\mu'_1=np$ and $\mu'_2=np(1-p+np)$.Thus, $\mu=n.p$ and

$\sigma^2=\mu'_2-(\mu'_1)^2=np\,(1-p+np)-(np)^2=n.p.(1-p)$

We can verify that the mean and the variance of binomial distribution are $\mu = np \text{ and } \sigma^2 = np(1-p),$ when n is the number of trials and p is probability of success and (1 – p) = q is the probability of failure as follows:.

$$\mu = \sum_{0}^{n} x \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \sum_{x=1}^{n} \frac{n!}{(x-1)!\,(n-x)!} p^x (1-p)^{n-x}$$

We have omitted the term x = 0 and cancelled the x against the first factor of $x! = x(x-1)!$ in the denominator of $\binom{n}{x}$. Then factoring out the factor $n! = n(n-1)!$ and one factor p we get

$$\mu = np \sum_{x=1}^{n} \binom{n-1}{x-1} p^{x-1} (1-p)^{n-x}$$

Letting y = x – 1 and m = n – 1, this becomes

$$\mu = np \sum_{x=1}^{n} \binom{m}{y} p^y (1-p)^{m-y} = np$$

The last summation is the sum of all the values of a binomial distribution with the parameters n and p and hence equal to 1.

To find the expression of $\mu'_2 \text{ and } \sigma^2$, let us make use of the fact that $EX^2 = E[X(X-1)] + E(X)$ and first evaluate $E[X(X-1)]$. Duplicating for all practical purposes the step used before, we get :

$$E[X(X-1)] = \sum_{x=0}^{n} x(x-1) \binom{n}{x} p^x (1-p)^{n-x}$$

$$= \sum_{x=2}^{n} \frac{n!}{(x-2)!\,(n-x)!} p^x (1-p)^{n-x}$$

$$= n(n-1)p^2 \sum_{x=2}^{n} \binom{n-2}{x-2} p^{x-2} (1-p)^{n-x}$$

Letting y = x – 2 and m = n – 2, this becomes

$$E[x(x-1)] = n(n-1)p^2 \sum_{x=0}^{m} \binom{m}{y} p^y (1-p)^{m-y}$$

$$= n(n-1)\sigma^2$$

Therefore $\mu'_2 = E[X(X-1)] + E(X) = n(n-1)\sigma^2 + np$

Finally, $\sigma^2 = \mu'_2 - \mu^2 = n(n-1)\sigma^2 + np + n^2p^2 = np(1-p)$

Example: Three coins are tossed. Find the probability of (i) 0 head, 1 head, 2 heads, 3 heads (ii) more than one head (iii) at least one head.

Ans. Let the random variable x is the number of heads obtained in a toss of three coins. If we denote the occurrence of head as success and the non-occurrence as failure and the coins are assumed to be unbiased then let

P = Probability of success in a single trial

= Probability of head in a single trial

= ½

Q = 1 – P = 1 – ½ = ½

n = Number of independent trials.

Since the value of P is constant and for each coin the trials are independent, the variable x follows a binomial distribution with parameters n = 3 and p = ½. Thus, probability of x success is $f(x) = {}_nC_x P^x Q^{n-x} = 3C_x(1/2)^x(1/2)^{3-x} = 3C_x(1/2)^3$

1) Putting the values of x = 0, 1, 2, 3 respectively

   f(0) = Probability of 0 head = $3C_0(1/2)^3 = 1 \times 1/8 = 1/8$

   f(1) = Probability of 1 head = $3C_1(1/2)^3 = 3 \times 1/8 = 3/8$

   f(2) = Probability of 2 heads = $3C_2(1/2)^3 = 3 \times 1/8 = 3/8$

   f(3) = Probability of 3 heads = $3C_3(1/2)^3 = 1 \times 1/8 = 1/8$

2) Prob. of more than one head = f(2) + f(3) = 3/8 +1/8 = 4/8 = ½

3) Prob. of at least one head = 1 – Prob (0 head) = 1 – f(0) = 1 – 1/8 = 7/8

Proposition: If x_1 follows a Binomial distribution with parameters (m_1, P) and x_2 follows a Binomial distribution with parameters (m_2, P), and x_1 and y_2 are statistically independent, then ($x_1 + x_2$) also follows Binomial distribution with parameters ($m_1 + m_2$, P).

The result can be extended to several independent variables with a common P.

Proof: Suppose x_1 and x_2 are distributed independently in the binomial form with parameters m_1, P and m_2, P respectively. Then the distribution of the sum ($x_1 + x_2$) can take values 0, 1, 2,…, $m_1 + m_2$.

Also, $$P[x_1, x_2 = k] = \sum_{k_1=0}^{k} P[x_1 = k_1]P[x_2 = k - k_1]$$

$$= \sum_{k_1=0}^{k} \binom{m_1}{k_1}\binom{m_2}{k-k_1} p^k(1-p)^{m_1+m_2-k}$$

$$= p^k(1-p)^{m_1+m_2-k} \sum_{k_1=0}^{k} \binom{m_1}{k_1}\binom{m_2}{k-k_1}$$

Now, this sum is nothing but the sum of products of the coefficients of t^{k-1} in $(1+t)^{m_1}$ and t^{k-k_1} in $(1+t)^{m_2}$ for varying k_1, the hence equals the coefficients of t^k in $(1+t)^{m_1+m_2}$, which is $\binom{m_1+m_2}{k}$

Thus, $P[x_1 + x_2 = k] = \binom{m_1+m_2}{k} p^k (1-p)^{m_1+m_2-k}$

This shows that $x_1 + x_2$ is itself binomially distributed with parameters $m_1 + m_2$ and p. We also get from the general result that if $x_1, x_2, ..., x_n$ are independently distributed binomial variables with parameters $m_1, p; m_2 p; ..., m_n, p;$ then the sum $x_1 + x_2 + ... + x_n$ is also a binomial variable with parameters $m_1 + m_2 + ... + m_n$ and p.

This implies that if $x_1, x_2, ..., x_n$ are a random sample from a binomial distribution with parameters of the statistic, $x_1 + x_2 + ... + x_n$ is also binomial with parameters n, m and p.

27.4.2 Poisson Distribution

When the number of trials discussed in the case of binomial distribution is very large, the calculation of the probabilities with binomial distribution becomes very cumbersome. Suppose we want to know what is the probability that in a clinical test 200 out of 300 mice will survive, after being infected by a virus. There is 50% chance that each mouse can fight with the virus by producing antibodies. If we use binomial distribution, the required probability is given by$^{300}C_{200}$.½ 300. In this section, we will discuss a probability distribution, which can be used to approximate binomial probabilities of this kind. Specifically, we will investigate the limiting form of the binomial distribution when n→∝ and p→0, while n.p remains constant. Let us suppose λ=n.p, therefore p=λ/n.

We can also write the binomial distribution as

$b(x,n,p) = {}^nC_x (\lambda/n)^x (1-(\lambda/n))^{n-x}$

$= n(n-1)(n-2)(n-3)\ldots(n-x+1)/ x! \times (\lambda/n)^x (1-\lambda/n)^{n-x}$

$= 1.(1-1/n)(1-2/n)(1-3/n)...(1-(x-1)/n)/x! \times \lambda^x \times [(1-(\lambda/n)) -^{n/\lambda}] -^{\lambda} \times (1-(\lambda/n))^{n-x}$

Finally, if we have n→∞ while x and p are constants

$$(1-1/n)(1-2/n)(1-3/n)...(1-(x-1)/n) \to 1$$

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$$(1-(\lambda/n)) -^{x} \to 1$$

$$(1-(\lambda/n)) -^{n/\lambda} \to e$$

Therefore, the limiting form of the binomial distribution becomes $P(x,\lambda) = \lambda^x e^{-\lambda} / x!$ for x= 0, 1, 2,3, ……….which is the p.m.f of a poisson random variable

Thus, in the limit when n → ∞, p →0 and np = λ remains constant; the number of success is a random variable which follows a poisson distribution with only one parameter λ. This distribution is named after the French mathematician Simeon Denis Poisson.

Following random variables are classic examples of poisson distribution.

1) Number of accidents on a road crossing.

2) Number of defects per unit area of a sheet material.

3) Number of telephone calls received by a telephone attendant.

4) Number of suicides in a year in an area etc.

Poisson distribution is a discrete probability distribution and is defined by the p.m.f: $f(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}; x = 0,1,2,\ldots,\infty;$

where λ is positive and is known as the parameter of Poisson distribution and e is the mathematical constant which is equal to 2.718 approx.

There are few important properties of the poisson distribution.

- Poisson distribution is a discrete probability distribution, where the random variable assumes countably infinite number of values such as 0,1,2,3,…….to ∞.The distribution is completely specified if the parameter λ is known.
- Mean and variance of poisson distribution are the same, both being λ.
- Poisson distribution, like the binomial distribution, may have either one or two modes. When λ is not an integer, mode is the largest value contained in λ and when λ is an integer, there are two modes, λ and $(\lambda-1)$
- The distribution has skewness = $1/\sqrt{\lambda}$ and kurtosis = $1/\lambda$. Therefore, we can conclude that poisson distribution is positively skewed and leptokurtic.
- If X_1 and X_2 are two random variables following the poisson distribution with parameters λ_1 and λ_2 respectively, then the random variable $(X_1 + X_2)$ also follows poisson distribution with the parameter $(\lambda_1+\lambda_2)$.
- As we have discussed earlier, the poisson distribution could be used as an approximation to binomial distribution when n is large but np is fixed.
- The moment generating function of the poisson distribution is given by $M_x(t) = E(e^{tX}) = e^{\lambda(e^t-1)}$
- $M_x(t) = \sum_{i=0}^{\infty} e^{tx}.f(x)$

$$= \sum_{i=0}^{\infty} e^{tx}.\lambda^{x} e^{-\lambda} / x!$$

$$= e^{-\lambda} \sum_{i=0}^{\infty} e^{tx}.\lambda^{x} / x!$$

$$= e^{-\lambda} \sum_{i=0}^{\infty} (\lambda e^t)^x / x!$$

In the above expression $\sum_{i=0}^{\infty} (\lambda e^t)^x / x!$ can be recognized as the Maclaurin's series of e^z where $z = \lambda e^t$.

Thus, the moment generating function of the Poisson distribution is

$$M_x(t) = e^{-\lambda}.e^{\lambda et}$$

$$M_x''(t) = \lambda.e^t.e^{\lambda(et-1)} + \lambda^2.e^{2t}.e^{\lambda(et-1)}$$

Therefore, $\mu_1' = M_x'(0) = \lambda$ and $\mu_2' = M_x''(0) = \lambda + \lambda^2$ and we get $\mu = \lambda$ and $\sigma^2 = \mu'_2 - (\mu'_1)^2 = \lambda + \lambda^2 - \lambda^2 = \lambda$

Example: Let X be a random variable following poisson distribution. If P(X=1) = P(X= 2), find P(X = 0 or 1) and E(X).

For poisson distribution, the p.m.f. is given by $P(x,\lambda) = \lambda^x e^{-\lambda}/x!$

Therefore, $P(X=1) = \lambda^1 e^{-\lambda}/1! = \lambda e^{-\lambda}$

$P(X=2) = \lambda^2 e^{-\lambda}/2! = \lambda^2 e^{-\lambda}/2$

As P(X=1) = P(X=2), from the equation $\lambda e^{-\lambda} = \lambda^2 e^{-\lambda}/2$, we get $\lambda=2$.

Therefore, $E(X) = \lambda = 2$ and

$P(X=0 \text{ or } 1) = P(X=0)+P(X=1) = 2^0 e^{-2}/0! + 2^1 e^{-2}/1! = 3.e^{-2}$.

Example: A random variable x follows a Poisson distribution with parameter 3. Find the probabilities that the variable assume the values: (i) 0, 1, 2, 3 (ii) less than 3 (iii) at least 2. Given that $e^{-3} = .0498$.

Ans. For the Poisson distribution with parameter $\lambda = 3$, the probability of x success is:

$$f(x) = \frac{e^{-3} 3^x}{x!}$$

i) Putting x = 0, 1, 2, 3 successively the required probabilities are:

$$f(0) = \frac{e^{-3} 3^0}{0!} = e^{-3} = .0498$$

$$f(1) = \frac{e^{-3} 3^1}{1!} = 3e^{-3} = 3(0.0498)$$

$$f(2) = \frac{e^{-3} 3^2}{2!} = \frac{9(0.0498)}{2}$$

$$f(3) = \frac{e^{-3} 3^3}{3!} = \frac{27(0.0498)}{6}$$

ii) Probability less than 3 success is the sum of the probabilities: f(0) + f(1) + f(2)

iii) Probability of at least two success:

$$f(2)+f(3)+f(4)+f(5)+...$$
$$=1-\{f(0)+f(1)\}=1-e^{-3}-3e^{-3}=1-4(0.0498)$$

Example: In a textile mill, on an average, there are 5 defects per 10 square feet of cloth produced. If we assume a poisson distribution, what is the probability that a 15 square feet cloth will have atleast 6 defects?

Let X be a random variable denoting the number of defects in a 15 square feet piece of cloth.

Since on an average, there are 5 defects per 10 square feet of cloth, there will be on an average 7.5 defects per 15 square feet of cloth i.e., $\lambda = 7.5$. We are to find $P(X\geq 6)=1-P(X\leq 5)$.

With the help of Table 27.5, we obtain $P(X \leq 5)=.2415$.

Therefore, $P(X\geq 6) = 1-.2415=.7585$.

Table 27.5: Probability distribution for number of defects

X	P (X)
0	0.0006
1	0.0041
2	0.0156
3	0.0389
4	0.0729
5	0.1094

Proposition: If x_1 and x_2 are independent poisson distributions with parameters λ_1, λ_2, respectively, then (x_1+x_2) also follows poisson distribution with parameters $(\lambda_1+\lambda_2)$.

Suppose x_1 and x_2 are distributed independently in the Poisson form with parameter λ_1 and λ_2 respectively. The sum x_1+x_2 can then take the values 0, 1, 2, …

Also, $P[x_1+x_2=k]=\sum_{k_1=0}^{k} P[x_1=k_1]P[x_2=k-k_1]$

$$=\sum_{k_1=0}^{k}\frac{\exp[-\lambda_1]\lambda_1^{k_1}}{k_1!}\times\frac{\exp[-\lambda_2]\lambda_2^{k-k_1}}{(k-k_1)!}$$

$$=\frac{\exp[-\lambda_1-\lambda_2]}{k!}\sum_{k_1=0}^{k}\binom{k}{k_1}\lambda_1^{k_1}\lambda_2^{k-k_1}$$

$$=\exp[-(\lambda_1+\lambda_2)]\frac{(\lambda_1+\lambda_2)}{k!},$$

which shows that $x_1 + x_2$ is itself a poisson variable with parameter $\lambda_1 + \lambda_2$. It immediately follows that if $x_1, x_2, ..., x_n$ are independently distributed poisson variables with $\lambda_1, \lambda_2, ..., \lambda_n$ then the sum $x_1 + x_2 + .. + x_n$ is also a Poisson variable with parameter $\lambda_1 + \lambda_2 + ... + \lambda_n$.

The above results give, in particular, the sampling distribution of the statistic $x_1 + x_2 + ... + x_n$ when $x_1, x_2, ... x_n$ are a random sample from a Poisson distribution with parameter λ. This sampling distribution is also of the Poisson form with parameter $n\lambda$.

Check Your Progress 2

1) The mean and standard deviation of a binomial distribution is given by 4 and $\sqrt{\left(\frac{8}{3}\right)}$ respectively. Find the values of n and p.

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2) Prove that poisson distribution is a limiting case of binomial distribution.

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3) The overall percentage of failure in a certain examination is 40. What is the probability that out of a group of 6 candidates at least 4 passed examinations?

   Hint: x follows a Binomial distribution and the event of a candidate passing the examination is a success. Thus, the probability of success: $f(x) = 6C_x(60/100)^x (40/100)^{6-x}$. Find out f(4) + f(5) + f(6)

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4) In 10 independent throws of a dice, the probability that an even number will appear 5 times is twice the probability that the even number will appear 4 times. Find the probability that an even number will not appear at all in 10 independent throws of the dice.

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5) In turning out some toys in a manufacturing process in a factory, the average number of defectives is 10%. What is the probability of getting exactly 3 defectives in a sample of 10 toys chosen at random ? Use poisson approximation to the binomial distribution (take e=2.72).

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6) 2% of the items made by a machine are defective. Find the probability that 3 or more items are defective in a sample of 100 items? (e-1 = 0.368, e-2 =.135, e-3=.0498)

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7) Find the probability that 7 out of 10 persons will recover from a tropical disease if we can assume independence and the probability that anyone of them will recover from the disease is 0.8.

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27.4.3 Normal Distribution

The normal distribution, which we will study in this section, is the corner stone of modern statistical theory. It was first investigated in the eighteenth century when scientists observed an astonishing degree of regularity in errors of measurement. They found that the patterns of the errors of measurement could be closely approximated by continuous curves, which they referred as the "normal curves". The mathematical properties of such continuous curves were first studied by Abraham de Moivre, Pierre Laplace and Karl Gauss.

A continuous random variable X is said to follow a normal distribution, and it is referred to as a normal random variable, if and only if its probability density is given by

$$f(x) = n(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\Pi}} \times e^{-\frac{1}{2}\{(x-\mu)/\sigma\}^2} \text{ for } -\infty < x < \infty, \text{ where } \sigma > 0$$

The shape of normal distribution is like a bell and is shown in Fig 27.2.

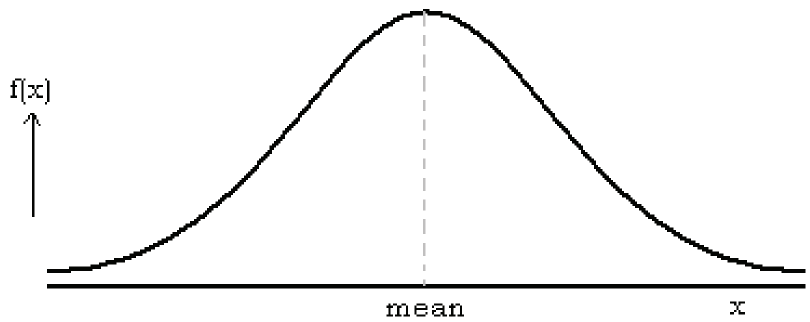

Fig. 27.2: Normal Distribution

While defining the p.d.f. of normal distribution we have used the standard notations, where σ stands for standard deviation and μ for mean of the random variable X. Note that f(x) is positive as long as σ is positive which is guaranteed by the fact that standard deviation of a random variable is always positive. Since f(x) is a p.d.f. while X can assume any real value, integrating f(x) over -∝ to ∝ we should get the value 1. In other words, the area under the curve must be equal to 1. Let us show that

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\Pi}} \times e^{\frac{1}{2}\{(x-\mu)/\sigma\}^2} \, dx = 1$$

We substitute $(x-\mu/\sigma = z$ in the R.H.S of the above equation to get

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\Pi}} \times e^{\frac{1}{2}\{(x-\mu)/\sigma\}^2} \, dx = \frac{1}{\sqrt{2\Pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} \, dz = \frac{2}{\sqrt{2\Pi}} \int_{0}^{\infty} e^{\frac{-1}{2}z^2} \, dz$$

[Since the p.d.f. is symmetrical, integrating the function from 0 to ∞ twice is the same as integrating the same function $-\infty$ to ∞]

= 2/√2∏×√∏/√2

= 1

Normal distribution has many nice properties, which makes it amenable to be applied to many statistical as well as economic models.

Properties of Normal Distribution

- Normal distribution is a continuous probability distribution.
- Normal distribution has two parameters, mean and standard deviation given by μ and σ respectively.
- Normal distribution is symmetric around the mean μ, which is at the same time the mode and the median of the distribution
- As a corollary to the above property, we can say that the first and third quartiles are equidistant from the mean of the normal distribution. Approximately,

  $Q_1 = \mu - 0.67 \times \sigma$
  and

  $Q_3 = \mu + 0.67 \times \sigma$
- All odd order central moments of the normal distribution are 0.

  In general $\mu_{2r} = 0$ for r = 1,2,3,…… μ_{2r-1}
- The two tails of the distribution are extended to infinity on both sides of the mean. The tails of the distribution never meet the horizontal axis. The maximum ordinate of the p.d.f is at the mean, which is given by 1/σ√2Π.
- However, beyond the points $x = \mu - 3\sigma$ and $x = \mu + 3\sigma$ the curve is close to the axis, that is the area under the curve beyond the two points in both

directions can be taken as zero.

- The points of inflexion (where the second derivate of f(x) is zero and changes sign), of the normal curve are located one standard deviation away from the mean, namely at $x = \mu + \sigma$ and $x = \mu - \sigma$ respectively. At these two points, normal curve changes its curvature.
- It is unimodal: its first derivative is positive for $x < \mu$, negative for $x > \mu$, and zero only at $x = \mu$.
- The probability that a normal random variable X equals any particular value is 0.
- The probability that X is greater than *a* equals the area under the normal curve bounded by *a* and plus infinity.
- The probability that X is less than *a* equals the area under the normal curve bounded by *a* and minus infinity.
- Additionally, every normal curve (regardless of its mean or standard deviation) conforms to the following "rule".
  1) About 68% of the area under the curve falls within 1 standard deviation of the mean.
  2) About 95% of the area under the curve falls within 2 standard deviations of the mean.
  3) About 99.7% of the area under the curve falls within 3 standard deviations of the mean.
- If a random variable X follows normal distribution with the mean and variance μ and σ respectively, then the random variable $z = (x - \mu)/\sigma$ is called standard normal variable. It has a density function:

$$f(x) = \frac{1}{\sqrt{2\Pi}} e^{\frac{-z^2}{2}} dz.....................-\propto < z < \propto$$

- The continuous probability distribution defined above is known as standard normal distribution. In fact, this is a special kind of probability distribution with mean zero and standard deviation 1.
- If X and Y are two normal variables with mean μ_1 and μ_2 and standard deviation σ_1 and σ_2, then (X + Y) is also a normal variable with mean $(\mu_1+\mu_2)$ and variance $(\sigma_1^2+\sigma_2^2)$.
- The moment generating function of a normal curve is given by

$$M_x(t) = e^{\mu t+1/2.\sigma^2 t^2}$$

$$M_x(t) = \int_{-\infty}^{\infty} e^{tx} \times \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

The above expression could be written, after some algebraic manipulation as the following:

$$M_X(t) = e^{\mu t+1/2(t\sigma)2} \times \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-1/2\left[\frac{x-(\theta+t\sigma^2)}{\sigma}\right]} dx = e^{\mu t+1/2(t\sigma)2}.$$

[since $1/\sigma\sqrt{2}\prod \int_{-\infty}^{\infty} e^{-1/2[\{x-(\mu+t\sigma 2)\}/\sigma]^2} dx = 1]$

Differentiating $M_X(t)$ with respect to t twice, we can get

$M'_X(t) = (\mu + \sigma^2 t)M_X(t)$

$M''_X(t) = [(\mu + \sigma^2 t)^2 + \sigma^2]M_X(t)$

Substituting t = 0 in the above two equations

$M'_X(0) = \mu$

$M''_X(0) = \mu^2 + \sigma^2$. Therefore, $E(X) = \mu$ and Variance $(X) = \sigma^2$.

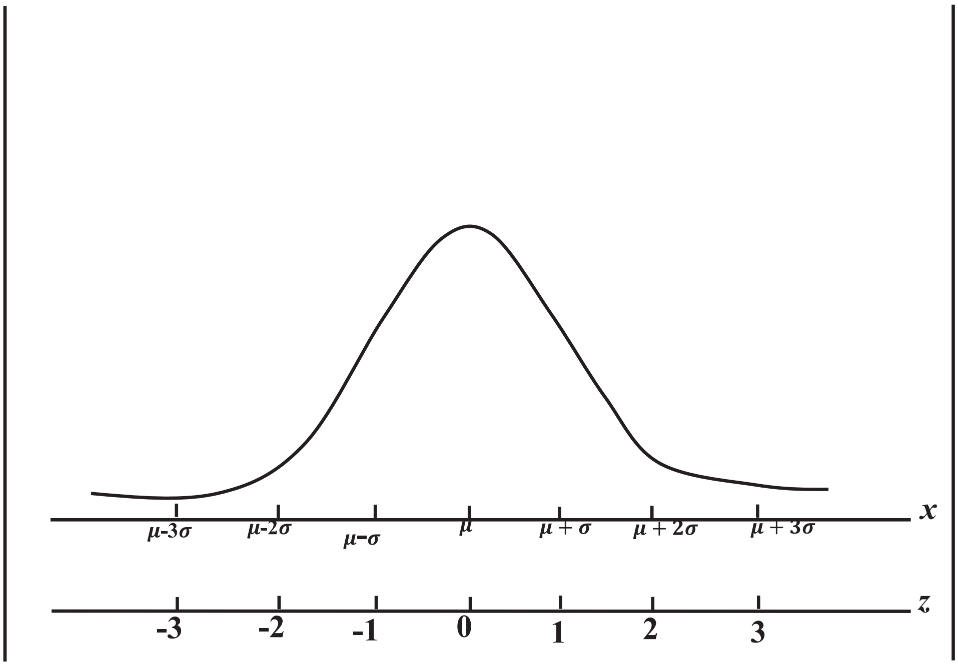

Fig. 27.3: Normal Distribution with Deviations from Mean

It x follows a normal distribution with mean as $\mu,$ and standard deviation $\sigma : x \sim N(\mu,\sigma)$ and z is another variate which is related to x such that $z = \frac{x-\mu}{\sigma}$. Then z follows a normal distribution $z \sim N(0,1)$ with 0 mean and 1 standard deviation. This simplest case of a normal distribution is known as the Standard Normal distribution. This is a special case when $\mu = 0 \text{and} \sigma = 1$ (Refer Fig 27.3). If $z = \frac{x-\mu}{\sigma}$ then $N(0,1) = f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}; -\propto \leq z \leq +\propto$

Following results regarding the area under the Standard Normal distribution are useful:

Area between $z = \pm 3$ is 99.73%

Area between $z = \pm 2.58$ is 99 %

Area between $z = \pm 1.96$ is 95%

Let the ordinate of F(p) be denoted by z_p, the value of the standard normal variate z such that the probability of its being exceeded is p i.e., $p(z > z_p) = p$. Area under the curve to the right of the ordinate at z_p is p.

Area of the right of $z = 1.645$ is $5\% = z_{.05}$

Area of the right of $z = 1.96$ is $2.5\% = z_{.025}$

Area of the right of $z = 2.33$ is $1\% = z_{.01}$

The approximate area under the standard normal curve is shown in Fig. 27.4.

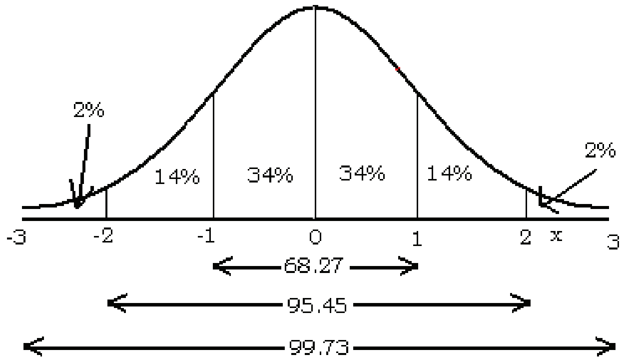

Fig 27.4: Area under the Standard Normal Curve

Example: The height distribution of a group of 10,000 men is normal with mean height 64.5″ and standard deviation 4.5. Find the number of men whose height is

a) less than 69″ but greater than 55.5″

b) less than 55.5″

c) more than73.5″

The mean and standard deviation of a normal distribution is given by

$\mu = 64.5''$ and $\sigma = 4.5$. We explain the problem graphically. As per Fig. 27.5, we can easily comprehend what we are asked to do. We are to find out the shaded regions but as we know area under a standard normal curve only, we have to reduce the given random variable into a standard normal variable.

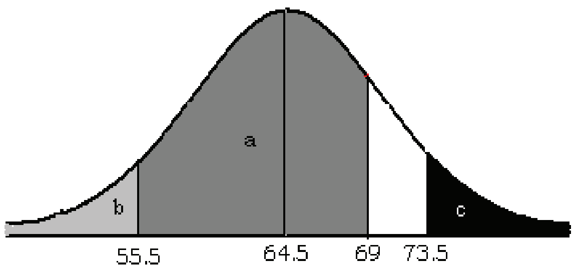

Fig 27.5: Standard Normal Distribution when μ = 64.5" and σ = 4.5.

Let X be the continuous random variable measuring the height of each man.

Therefore, (X–64.5)/4.5 = z is a standard normal variable.

Table 27.6 shows values of z for corresponding values of x. In the table, true area under standard normal curve is given for only positive values of the standard normal variable and as the distribution is symmetrical, area under the curve for negative values of the standard normal variable is easy to find out.

Table 27.6: Values of z as per x value

X	Z
55.5	–2
64.5	0
69	1
73.5	2

a) $P(55.5 < X < 69) = P(-2 < Z < 1) = \Phi(1) - \Phi(-2) = .82$, Therefore, men of height less than 69″ but greater than 55.5″ is 10000 × 0.82 = 8200

b) $P(X < 55.5) = P(z < -2) = .02$, Therefore, men of height less than 55.5" is 10000 × 0.02 = 200

c) $P(X > 73.5) = P(z > 2) = 1 - P(z < 2) = 1 - .98 = .02$, Therefore, men of height greater than 73.5″ is 10000× 0.02 = 200

For standard normal curve (say for the variable z) the area under the curve to the left is conventionally denoted by $\Phi(z_1)$. This is shown in Fig. 27.6.

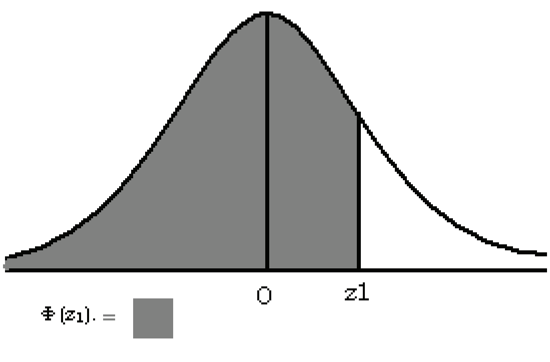

Fig 27.6: Area under the Standard Normal Curve denoted by $\Phi(z_1)$

Distribution of Standard Normal Variable

A standard Normal variable of a normal deviate is generally denoted by τ (or z), so that the p.d.f. of standard normal distribution may be written as

$$f(\tau) = \frac{1}{\sqrt{2\pi}} \exp(-\tau^2/2), -\alpha < \tau < \alpha$$

The distribution is symmetrical about 0. The p.d.f of standard normal distribution is

$$f(\tau) = \frac{1}{\sqrt{2\pi}} \exp(-\tau^2/2), -\alpha < \tau < \alpha$$

The properties of this distribution may be deduced form those of a general normal distribution. We shall denote by τ_α the value of τ such that

$$p[\tau > \tau_\alpha] = \alpha$$

It is called the upper $\alpha -$ point (or the upper 100 $\alpha\%$ point) of a standard variable. Because of the symmetry of the distribution about zero, we have

$$\tau_{1-\alpha^-} = -\tau_\alpha$$

Thus, the lower α point, $\tau_{1-\alpha}$ of a standard normal variable is the value of τ such that $p[\tau > \tau_{1-\alpha}] = \alpha$ which is the same as the upper α point in magnitude but has the opposite sign.

It follows from the theorem below that if x is normally distributed with mean μ and variance σ^2, then $(x-\mu)/\sigma$ is a standard normal variable. Conversely, if $(x-\mu)/\sigma$ is a standard normal variable, then x is a normal variable with mean μ and variance σ^2.

Theorem: If x is normally distributed with mean μ and variance σ^2, then $y = a + bx$, where $b \neq 0$, is also normally distributed with mean $a + b\mu$ and variance $b^2\sigma^2$

Proof: Let us denote the p.d.fs of x and y by f(x) and g(y), respectively.

Assuming **b > 0,** from the result if y = a + bx, then $\frac{y-a}{b} = x$

$$p[c < y < d] = p\left[\frac{c-a}{b} < x < \frac{d-a}{b}\right],$$

$$\int_c^d g(y)\,dy = \int_{(c-a)/b}^{(d-a)b} f(x)dx = \int_c^d f\left(\frac{y-a}{b}\right)\frac{dx}{dy}.dy \qquad \ldots\ldots\ldots\ldots\ldots(1)$$

If **b < 0**, we similarly have,

$$p[c < y < d] = p\left[\frac{d-a}{b} < x < \frac{c-a}{b}\right]$$

$$\int_c^d g(y)\,dy = -\int_c^d f\left(\frac{y-a}{b}\right)\frac{dx}{dy}dy \qquad \ldots\ldots\ldots\ldots\ldots(2)$$

Combining the two results (1 & 2), we get

$$g(y) = f\left(\frac{y-a}{b}\right)\left|\frac{dx}{dy}\right|$$

Note that

$$\frac{dx}{dy} = 1\Big/\left(\frac{dy}{dx}\right) = \frac{1}{b}$$

But

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left[-(x-\mu)^2\Big/2\sigma^2\right]$$

and $$g(y) = \frac{1}{|b|\sigma\sqrt{2\pi}}\exp\left[-\left(\frac{y-a}{b}-\mu\right)^2\Big/2\sigma^2\right]$$

Hence

$$= \frac{1}{|b|\sigma\sqrt{2\pi}}\exp\left[-(y-a-b\mu)^2\Big/2b^2\sigma^2\right].$$

Check Your Progress 3

1) The mean weight of 500 students at a university is 151 lbs and the standard deviation is 15. Assuming the weights are normally distributed, find how many students:

   i) Weigh between 120 and 155 lbs.

   ii) Weigh Morethan155lbs.

   Given: Φ(0.27) = 0.6064; Φ(2.07) = 0.9808; Φ(t) implies the area under the standard normal curve to the left of the ordinate at the point t.

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2) The mean of a normal distribution is 50 and 5% of the values are greater than 60. Find out the standard deviation of the distribution (given that the area under the standard normal curve between z =0 and z = 1.64 is 0.45)

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27.5 LET US SUM UP

In this unit, you learnt the concepts of joint distribution and the associated marginal distribution. You have also been introduced to the three most elementary and most used distributions in the theory of probability namely, Binomial, Poisson and Normal. First two of these are discrete and the last one is continuous. A distribution is defined mostly by its moments; therefore, having understood the concepts concerning moments we discussed how these moments are used to characterize a distribution.

27.6 KEY WORDS

Binomial Distribution : A discrete probability distribution satisfying the following conditions is a binomial distribution:

1) It involves finite repetition of identical trials.
2) Each trial has two possible outcomes: success and failure.
3) Trials are independent of each other.
4) The probability of the outcomes (success and failure) does not change from one trial to another.

Normal Distribution: : It is a continuous probability distribution with the following properties:

1) It is symmetrical about its mean;
2) All the measures of central tendency for this distribution is the same, i.e., for normal distribution Mean = Mode = Median; and
3) A variable following normal distribution can take any value within the range $(-\infty, \infty)$

Poisson Distribution : A discrete probability distribution which is the limiting form of the binomial distribution, provided

1) The number of trials is very large is fact tending to infinity.
2) The probability of success in each trial is very small; tending to zero.

Probability Distribution : Probability distribution of a random variable is a statement specifying the set of its possible values together with the respective probabilities.

27.7 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) $f_X(x) = \int_0^{3-3x} \frac{x+y}{4} dy = \frac{9}{8} - (3/2)x + \frac{3}{8}x^2, 0 < x < 1$

$f_Y(y) = \int_0^{1-(y/3)} \frac{x+y}{4} dx = \frac{1}{8} + \frac{1}{6}y - \frac{5}{72}y^2, 0 < y < 3$

2) Hint: $g(x) = g(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_2^1 \frac{2}{3}(x+2y)\,dy = \frac{2}{3}(x+1)$ for $0 < x < 1$ and $g(x) = 0$ elsewhere

Likewise, you can find out h(y)

Check Your Progress 2

1) As the formula for mean and variance of binomial distribution are given by n.p and $n.p.(1-p)$ respectively, we get the following two equations

n.p= 4… .... (1)

n.p.(1-p) = 8/3… .... (2)

Solving these two equations we get n = 12 and p = 1/3.

2) See Section 27.4.2.

3) Hints: x follows a Binomial distribution and the event of a candidate passing the examination as success. Thus the probability of success: $f(x) = 6C_x (60/100)^x (40/100)^{6-x}$. Find out f(4) + f(5) + f(6)

Ans: 1701/3125

4) Hints: Let the $f(x) = 10C_x (P)^x (Q)^{10-x}$: and it is given that f(5) = 2f(4)

Ans: $\left(\frac{3}{8}\right)^{10}$

5) λ =10 × 0.1 = 1,hence the probability of three defectives in the sample is given by

$f(3)=e^{-1}\times 1^3/3!=0.061$

6) The number of defectives follows a binomial distribution. Since p= .02 and n=100 which is very large, making up, n.p = λ = 100×.02 = 2, a finite quantity, we use the poisson approximation to binomial distribution.

P (of 3 or more defectives)=1–[f(0)+f(1)+f(2)]=1–$e^{-2}[1+2+2^2/2!]$= 0.325

7) Substituting x= 7, n = 10, and p = 0.8 into the formula for binomial distribution, we get

$b(7;10,0.8) = {}^{10}C_7.(0.8)^7(0.2)^3 = 0.2$ (approximately)

Check Your Progress 3

1) Clearly, the random variable denoting the weight, say X, of the students follow a normal distribution, mean (X) = 151; Var (X)= 15^2

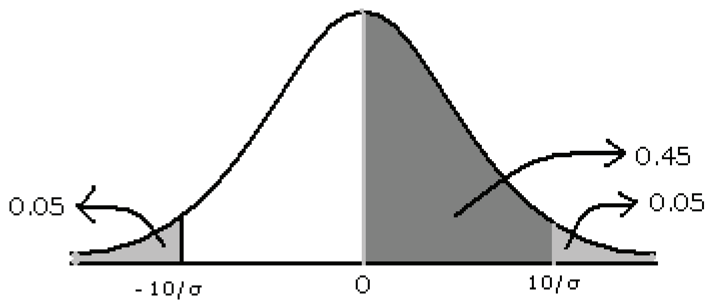
i) Proportion of students whose weight lie between 120 and 155 lbs =Area under the standard normal curve between the vertical lines at the standardized values; z = (120–151)/15= –2.07 and z = (155-151)/15 =0.27

$P(120 \le X \le 155) = \Phi(0.27)–\Phi(–2.07) = \Phi(0.27)–\{1–\Phi(2.07)\}$

$=0.6064–1+0.9808=0.5872$

ii) $P(X >155) =1– P(X \le 155) =1–\Phi (0.27) = 0.3936$

2) The probability that X takes values greater than 60 is 5% or 0.05. This must be the area under the standard normal curve to the right of the ordinate at the standardized value z = (60–50)/σ= 10/σ



Since the area to the right of z = 0 is 0.05 and the area between z = 0 and z =1.64 is given to be 0.45, the area to the right of z =1.64 is .5 –.45=.05. Thus, we get 10/σ = 1.64, or, σ = 6.1.

27.8 EXERCISES

Q1. Find the binomial distribution of getting a six in three tosses of an unbiased dice.

Ans: Let X be the random variable of getting six. Then X can be 0, 1, 2, 3.

Here, n = 3

p = Probability of getting a six in a toss = ⅙

q = Probability of not getting a six in a toss = 1 – ⅙ = ⅚

$P(X = 0) = {}^nC_r\, p^r\, q^{(n-r)} = {}^3C_0\, (1/6)^0\, (5/6)^{3-0}$

$= 1 \times 1 \times 125/216 = 125/216$

$P(X = 1) = {}^nC_r\, p^r\, q^{(n-r)} = {}^3C_1\, (1/6)^1\, (5/6)^{3-1}$

$= 3 \times 1/6 \times 25/36 = 25/72$

$P(X = 2) = {}^nC_r\, p^r\, q^{(n-r)} = {}^3C_2\, (1/6)^2\, (5/6)^{3-2}$

$= 3 \times 1/36 \times 5/6 = 5/72$

$P(X = 3) = {}^nC_r\, p^r\, q^{(n-r)} = {}^3C_3\, (1/6)^3\, (5/6)^{3-3}$

$= 1 \times 1/216 \times 1 = 1/216$

The required binomial distribution of X is:

X	0	1	2	3
P(X)	125/216	25/72	5/72	1/216

Q2. There are four fused bulbs in a lot of 10 good bulbs. If three bulbs are drawn at random with replacement, find the probability distribution of the number of fused bulbs drawn.

Ans: This is a problem of binomial distribution as the event of drawing a fused bulb is independent.

p = P(drawing a fused bulb) = 4/(10 + 4) = 2/7

q = P(drawing a bulb which is not fused) = 1 – 2/7 = 5/7

X = event of drawing a fused bulb

X can take up the values 0, 1, 2, 3

P(X = 0) = P(getting zero fused bulbs in all draws)

$= {}^nC_r\, p^r\, q^{(n-r)}$

$= {}^3C_0\, (2/7)^0\, (5/7)^{(3-0)}$

$= 1 \times 1 \times (125/343) = 125/343$

P(X = 1) = P (getting one time fused bulb)

$= {}^nC_r\, p^r\, q^{(n-r)}$

$= {}^3C_1\, (2/7)^1\, (5/7)^{(3-1)}$

$= 3 \times (2/7) \times (25/49) = 150/343$

P(X = 2) = P(getting two times fused bulbs)

$= {}^nC_r\, p^r\, q^{(n-r)}$

$= {}^3C_2\, (2/7)^2\, (5/7)^{(3-2)}$

$= 3 \times 4/49 \times (5/7) = 60/343$

P(X = 3) = P(getting three times fused bulb)

$= {}^nC_r\, p^r\, q^{(n-r)}$

$= {}^3C_3\, (2/7)^3\, (5/7)^{(3-3)}$

$= 1 \times 8/343 \times 1 = 8/343$

The required probability distribution:

X	0	1	2	3
P(X)	125/343	150/343	60/343	8/343

Q3. A life insurance salesman sells on the average 33 life insurance policies per week. Use Poisson's law to calculate the probability that in a given week he will sell

a) Some policies

b) 2 or more policies but less than 5 policies.

c) Assuming that there are 5 working days per week, what is the probability that in a given day he will sell one policy?

Ans. Here, $\mu = 3$

a) "Some policies" means "1 or more policies". We can work this out by finding 1 minus the "zero policies" probability:

$P(X > 0) = 1 - P(x_0)$

Now $P(X)=e^{-\mu}\mu^x / x!$

So,$P(x_0) = e^{-3} 3^0/0!$

$=4.9787\times10^{-2}$

Therefore, the probability of 1 or more policies is given by: Probability = $P(X \geq 0)$

$=1-P(x_0)$

$=1-4.9787\times10^{-2}$

$=0.95021$

b) The probability of selling 2 or more, but less than 5 policies is: $P\ (2 \leq X < 5)$

$= P(x_2)+P(x_3)+P(x_4)$

$= (e^{-3}\ 3^2/2!) + (e^{-3}\ 3^3/3!) + (e^{-3}\ 3^4/4!)$

$=0.61611$

c) Average number of policies sold per day: 3/5=0.6

So, on a given day, $P(X)=e^{-0.6}(0.6)^1/1!$

$=0.32929$

Q4. For some computers, the time period between charges of the battery is normally distributed with a mean of 50 hours and a standard deviation of 15 hours. Rohan has one of these computers and needs to know the probability that the time period will be between 50 and 70 hours.

Ans: Let x be the random variable that represents the time period.

Given Mean, $\mu= 50$

and standard deviation, $\sigma = 15$

To find probability that x is between 50 and 70 or $P(50< x < 70)$:

By using the transformation equation, we know;

$z = (X - \mu) / \sigma$

For $x = 50$, $z = (50 - 50) / 15 = 0$

For $x = 70$, $z = (70 - 50) / 15 = 1.33$

$P(50 < x < 70) = P(0 < z < 1.33)$ = [area to the left of $z = 1.33$] – [area to the left of $z = 0$]

From the table we get the value, such as;

$P(0 < z < 1.33) = 0.9082 - 0.5 = 0.4082$

The probability that Rohan's computer has a time period between 50 and 70 hours is equal to 0.4082.

Q5. The speeds of cars are measured using a radar unit, on a motorway. The speeds are normally distributed with a mean of 90 km/hr and a standard deviation of 10 km/hr. What is the probability that a car selected at chance is moving at more than 100 km/hr?

Ans: Let the speed of cars be represented by a random variable 'x'.

Now, given mean, $\mu = 90$ and standard deviation, $\sigma = 10$.

To find probability that x is higher than 100 or $P(x > 100)$:

By using the transformation equation, we know;

$z = (X - \mu) / \sigma$

Hence,

For $x = 100$, $z = (100 - 90) / 10 = 1$

$P(x > 100) = P(z > 1)$ = [total area] – [area to the left of $z = 1$]

$P(z > 1) = 1 - 0.8413 = 0.1587$